

# Lame Duck by Primary: Effects of Electoral Incentives on U.S. House Representatives

## **Abstract**

How do electoral incentives motivate representatives to exert effort and shape policy positions? While reelection is widely recognized as representatives' primary motivation in democratic politics, accurately estimating its effects is challenging because representatives self-select into rerunning or retiring decisions. Leveraging legislators' lame duck status after primary elections, this article estimates how electoral incentives affect U.S. House Representatives. Compiling data from 1955 to 2022, I compare legislators who win primaries and remain motivated for reelection to those who lose and exit the race. Through regression discontinuity and difference-in-differences designs, I find that electoral incentives operate on one margin but not the other. Primary losers withdraw effort across the board: they sharply reduce roll-call participation and cut bill sponsorship, cosponsorship, and constituent communication by roughly a third to two-thirds of their own baseline rates. Yet neither winners nor losers adjust their policy positions: ideological and party-line adherence remain stable, and the estimates rule out a party-line shift of even two votes in a hundred. Electoral incentives govern how hard representatives work, not where they stand.

Keywords: Electoral accountability, electoral incentives, legislative politics, U.S. House Representative

# 1 Introduction

Does the pursuit of reelection prompt representatives to work harder or shift their policy stances? Numerous scholars have emphasized the critical role of electoral incentives in democratic systems, arguing that the desire to secure reelection motivates officeholders to remain closely aligned with their constituents' interests and expectations. Elected officials may be motivated by goals other than reelection such as advancing their preferred policy agendas or heightening their standing within political parties, but the prevailing view holds that “[the electoral goal] must be the *proximate* goal of everyone, the goal that must be achieved repeatedly if other ends are to be entertained” (Mayhew, 2004). In this way, a drive for reelection serves as a fundamental mechanism of democratic accountability, ensuring political leaders' responsiveness to voter preferences and setting democratic governance apart from autocratic rule.

Despite the importance of electoral incentives in democratic politics, however, existing research offers conflicting theoretical perspectives and mixed empirical findings on what specific behaviors these incentives elicit from representatives. In theory, electoral incentives could pull a representative's behavior in opposing directions, or fail to move it at all when other forces dominate. Representatives may strive to meet voters' demands to secure reelection (Mayhew, 2004; Fenno, 1978). At the same time, they may (or may not) feel pressure from party leaders to act otherwise as they control crucial campaign and career resources (Rohde, 2010; Aldrich, 2008; Cox and McCubbins, 2005; Krehbiel, 1993). Alternatively, if representatives hold strong personal ideological commitments, electoral incentives may induce little change in their behavior at all (Alesina, 1988; Osborne and Slivinski, 1996; Besley and Coate, 1997; Poole and Rosenthal, 1985).

A core challenge in empirically evaluating theoretical claims about the impact of electoral incentives lies in disentangling these incentives from legislators' strategic decisions about whether to seek reelection or retire. The effect of electoral incentives is a contrast within the same legislator: behavior with the prospect of facing voters again, and behavior without it.

Behavior without that prospect is observable only among incumbents who are leaving office, so empirical analyses compare the behavior of incumbents who run for another term with the behavior of those who step down. However, rerunning or retiring decisions are shaped by a range of personal and political factors, making it difficult to attribute observed behavioral differences solely to the presence or absence of reelection motives. As a result, much of the literature has relied on theoretical assumptions and correlational evidence rather than on research designs that separate the effect of electoral incentives from selection into leaving office. Previous scholars have addressed this selection bias through institutional levers—term limits, electoral reforms, party control of renomination. Yet each removes electoral incentives in a way that is either anticipated far in advance or itself a strategic party decision, and many elective offices, including the U.S. House, offer no such institutional lever at all, so robust evidence on the behavioral effects of electoral incentives has remained correspondingly limited.

This article addresses this empirical challenge in the context of the U.S. House of Representatives by exploiting the unique opportunity presented by incumbent legislators’ lame duck status after primary elections. By collecting and combining data from over 3,800 incumbent legislators and their primary election contests between 1956 and 2022, I compare the post-primary behavior of primary election winners, who remain motivated for the upcoming general election, with that of primary election losers, who are disqualified from running for reelection. To ensure comparability between winners and losers, I employ two complementary research designs. A regression discontinuity design compares incumbents who narrowly win or lose primary elections, and a difference-in-differences design contrasts legislators’ behavior across terms in which they lost or won renomination.

The findings from both designs provide consistent evidence that electoral incentives shape how much effort legislators invest. After losing primary elections, legislators sharply reduce their roll-call participation, and the timing of the decline points to the primary itself: winners and losers are nearly indistinguishable until the closing weeks of the campaign, but the gap

widens discretely once the primary determines who remains a candidate and who does not, and persists through the general election. The withdrawal is not confined to the roll-call votes that others schedule: bill sponsorship, cosponsorship, and e-newsletter communication with constituents all decline in step, each activity falling by roughly a third or more of a legislator's own rate. The e-newsletter decline follows the same post-primary pattern as roll-call participation.

The withdrawal tracks neither the competitiveness of the upcoming general election nor the length of the career at stake: even where that election is a foregone conclusion, primary winners keep up their effort and losers fall behind them as surely as in the closest races, and younger losers, with the longest careers still ahead of them, withdraw as sharply as those near retirement. What losing a primary removes is candidacy itself—having a next election to face—not the marginal return of campaigning or the value of a career still to protect.

On the other hand, the findings from both designs show that electoral incentives do not meaningfully affect legislators' policy positions. Both primary election winners and losers remain largely consistent in terms of ideological extremity and party-line adherence following primary elections, contrary to a long line of theory in which the positions legislators take respond to electoral pressure (e.g., Downs, 1965; Mayhew, 2004). This stability is not an artifact of imprecision: the estimates are sharp enough to detect a party-line shift as small as two votes in a hundred, yet none appears. To investigate the extent of this stability, I further restrict the analysis to bills that receive greater attention from party leaders, bills that scholars have marked as significant, and contested bills that are close to passing or failing. The results show that neither primary election winners nor losers significantly change their policy positions even for these bills, underscoring the substantive strength of the null findings.

These findings carry several implications. By separating the margin of legislative behavior that responds to electoral incentives (how much effort to supply) from the one that does not (what positions to take), they sharpen the foundational assumptions of electoral accountability: the effort results reinforce Mayhew's premise that reelection is the proximate

motivation (Mayhew, 2004), while the policy-position null cuts against the corollary that legislators’ roll-call positions are themselves electorally driven—and, more broadly, against the Downsian account of electoral competition (Downs, 1965). The null instead fits models in which legislators’ ideological stances are largely fixed (Alesina, 1988; Osborne and Slivinski, 1996). The stability of incumbent positions also bears on debates over party influence (Cox and McCubbins, 2007; Krehbiel, 1993), the sources of polarization, and the reliability of roll-call-based ideology measures: it suggests that polarization operates more through electoral replacement of legislators than through within-term repositioning (Barber and McCarty, 2015; McCarty, 2019; Moskowitz, Rogowski and Snyder, 2024), and that scores such as DW-NOMINATE (Poole and Rosenthal, 1985; McCarty, 2016) capture stable preferences rather than artifacts of immediate electoral pressure.

## 1.1 Theoretical Considerations on Legislators’ Behavior

To understand how electoral incentives affect legislators’ behavior, this article focuses on two key aspects: effort-related behavior and policy positions based on roll-call votes. Standard models of electoral accountability suggest that electoral incentives drive elected officials to increase their effort levels in hopes of securing reelection (Barro, 1973; Ferejohn, 1986). However, when representatives must allocate limited resources across multiple tasks, it is less clear which specific activities will be prioritized (Fenno, 1973; Holmstrom and Milgrom, 1991). For example, electoral incentives may push legislators to attend roll-call votes to avoid perceptions of shirking, while also pulling them away from such floor activity toward campaigning and district demands.

When it comes to legislators’ policy positions, theoretical predictions diverge. Some models predict electoral incentives pull candidates toward the median voter (Downs, 1965; Calvert, 1985), while others suggest legislators shift toward positions that mobilize core supporters (Adams and Merrill, 2003; Fiorina, 1974).

A different theoretical tradition, or citizen-candidate and commitment-based models,

instead assumes that legislators’ policy positions remain largely fixed: candidates are treated as ideologues who steadfastly adhere to their preferred ideological stances (Alesina, 1988; Osborne and Slivinski, 1996; Besley and Coate, 1997). According to this strand of models, electoral incentives would have no effect on legislators’ policy positions.

It also remains unclear how much parties influence their legislators through electoral incentives, given the estimation challenge of legislators’ selection into parties (Smith, 2007; Krehbiel, 1993; Cox and McCubbins, 2007). If party influence exists, legislators may adopt party-line stances when seeking reelection, since the party supplies campaign resources. Donors and interest groups may exert similar pressure, each steering legislators toward the positions it favors, though evidence of their influence is mixed (Stratmann, 2002; Ansolabehere, de Figueiredo and Snyder, 2003; Fowler, Garro and Spenkuch, 2020). Because these external pressures may pull in different directions, the net effect of electoral incentives on policy positions is unclear.

Comparing rerunning and leaving legislators is the most natural empirical strategy, as the former faces the prospect of an upcoming election while the latter does not. However, even setting aside selection differences between the two groups, its interpretation hinges on the motivations and incentives that drive not only the former but also the latter. Leaving legislators are often expected to reduce their legislative activity or coast through their remaining term, but alternative motivations can keep them active in office. For instance, they may anticipate post-Congress appointments to federal or local office, or prepare for careers in the private sector or academia. Personal policy preferences or ideological commitments may also keep them active; freed from electoral pressures, they may pursue these commitments more openly than before, so the comparison recovers the effect of electoral incentives only net of these substituted motivations. In particular, if leaving legislators themselves remain motivated by future electoral concerns—say, sustaining a record for a later bid—the comparison reflects only a difference in horizon, not the presence or absence of electoral incentives, and cannot identify their effect.

These alternative motivations complicate the interpretation of this comparison as a clean test of electoral incentives. The major complication depends on what causes leaving. Voluntary retirements and anticipated departures by term limits let alternative motivations work through anticipatory behavior, making it difficult to isolate the effects of electoral incentives. A less anticipated event like a primary defeat closes this anticipatory channel, but the comparison’s cleanness still depends on what motivates the unanticipated leavers themselves. Reading the rerunning-versus-leaving comparison as a clean test of electoral incentives therefore requires a closer look at leaving legislators’ motivations and trajectories, which the next subsection takes up as part of its review of previous research.

## 1.2 Previous Research on Electoral Incentives

Estimating the effects of electoral incentives is challenging because it requires a credible counterfactual for what a legislator would do without them. Previous research has approached this challenge through several distinct designs, each constructing the counterfactual in a different way.

Earlier literature on shirking examined whether legislators behave differently in their last term, when they are not seeking reelection (Lott, 1987; Lott and Davis, 1992; Zupan, 1990). Because leaving legislators do not face electoral incentives, they can serve as a counterfactual, but only if they are otherwise comparable to those seeking reelection. If retirement decisions are strategic, or if retirees differ in quality or ideology from those seeking reelection, simple comparisons would introduce selection bias. Because these studies relied on such comparisons, they could not separate the effect of electoral incentives from systematic differences in who leaves office.

Building on earlier research, scholars have attempted to address this selection bias through various research designs. These approaches include leveraging term limits (Alt, Bueno de Mesquita and Rose, 2011; Fourniaies and Hall, 2022; Besley and Case, 1995), electoral reforms (Motolinia, 2021), and parties’ candidate nomination processes (Fiva and Nedregård, 2024);

each relaxes the selection problem, but none applies to the individual reelection incentive of U.S. House incumbents.

Similar to this article, Fiva and Nedregård (2024) employ party renomination decisions in Norway to compare legislators’ within-term behavior following the renomination outcome. Both studies find that legislators who fail to secure renomination tend to reduce their effort-related activities while largely maintaining their policy positions. Aside from the comparability concerns inherent to party-led renomination decisions in Fiva’s setting, the two works are situated in markedly different political contexts. In Norway, party leaders decide which incumbent legislators are renominated, a process characterized by strong party discipline and limited personal campaigning. In contrast, in the United States, incumbents must compete in primary elections that emphasize individual candidacies. Consequently, while Fiva and Nedregård (2024) highlight the influence of party leadership in shaping legislators’ behavior, this article centers on how individual legislators pursue reelection under a more candidate-driven electoral system.

Using a term-limit design with state legislators, Fourinaies and Hall (2022) also report similar findings: termed-out legislators reduce productivity yet maintain policy positions. The key difference is electoral salience. U.S. House races attract far more public attention than state-level contests, and past work attributes the policy-position null among state legislators specifically to that low salience (Rogers, 2017, 2023). The finding that the same null obtains in the high-salience U.S. House context is therefore striking, contrary to claims from earlier studies (e.g., Ansolabehere, Snyder and Stewart, 2001; Griffin, 2006; Fiorina, 1973).

Compared to a term-limit framework, this article’s use of primary elections offers a more precise way to differentiate between legislators motivated by electoral considerations and those who are not. As most House Representatives exit electoral politics after a primary defeat (Table 1), comparing primary winners and losers provides a clearer gauge of the impact of electoral incentives on legislative behavior. By contrast, termed-out state legislators may

**Table 1:** Career Trajectory After Losing in Primary Election

| <b>Run for<br/>Elections Later</b> | <b>Non-elected<br/>Public Position</b> | <b>Non-public<br/>Career</b> | <b>Retire or<br/>No Records</b> |
|------------------------------------|----------------------------------------|------------------------------|---------------------------------|
| 28<br>(17.50%)                     | 27<br>(16.88%)                         | 63<br>(39.38%)               | 42<br>(26.25%)                  |

*Note:* The table shows the distribution of House representatives’ post-Congress career trajectories after losing in primary elections. Only 17.5% of them ended up running for an electoral position again, while the majority pursued non-electoral or non-public roles, retired, or left no trace of their activities after leaving Congress.

still respond to future political prospects and party pressure: Steen (2006) shows that those hitting state-legislative term limits are more likely to run for Congress than those still eligible to serve, Fourniaies and Hall (2022) find stronger productivity effects in states where term limits prohibit running for the other chamber, and Olson and Rogowski (2020) document continuing party pressure on termed-out legislators. These ongoing incentives confound any attempt to isolate the effect of reelection pressure.

Other studies compare general-election winners and losers in brief November–December postelection lame-duck sessions (Jenkins and Nokken, 2008*a,b*; Schönenberger, 2024), and they face their own limitations. Such sessions see little new legislative activity since bills introduced then expire with the outgoing Congress, and what business does occur is largely confined to must-pass items—appropriations, the National Defense Authorization Act, and continuing resolutions—shaped by strategic party agenda-setting. Primary losers, by contrast, are observed over the months between the primary and the general election, when the House conducts its ordinary business and member-initiated activity continues, not in a truncated postelection window. Furthermore, general-election winners with a two-year horizon are less immediately motivated by reelection than primary winners, who face a general election within months and provide a sharper baseline.

Finally, Cowburn and Sältzer (2025) employ a research design most similar to this article’s by comparing winners and losers of primary elections among U.S. House representatives. They analyze legislators’ language usage on X (Twitter) in 2020 and find that Democratic

primary losers took significantly more moderate positions after their defeat, whereas no such effects were observed among Republicans. By contrast, this article examines a wider array of legislative behavior, focusing on various effort-related activities and policy positions based on roll-call votes, over a more extensive period from 1955 to 2022. The outcome constructs also differ: their measure is positioning expressed in campaign rhetoric, which legislators can recast freely, while this article’s is the roll-call record itself—a difference that can reconcile their post-defeat moderation with the position stability documented below.

## 2 Data on Primary Elections and Legislator Behavior

In the United States, primary elections typically occur several months before the November general election—as early as March and as late as September—providing parties an opportunity to select their preferred candidate to compete for the office. For incumbent legislators, these contests are especially significant: a primary win maintains their eligibility and drive to compete in the general election, while a defeat effectively disqualifies them from running for reelection.

This article exploits these features of primary elections by comparing the post-primary behavior of incumbent House representatives who win their primary contests with that of those who lose.

Pettigrew, Owen and Wanless (2014) provide primary election results for U.S. House Representatives from 1956 to 2010, and Miller and Camberg (2020) provide them from 2012 to 2018. These data do not include runoff elections, so I manually collected them from state archives and past newspapers; I also collected results for 2020 and 2022 from state archives. The combined data cover U.S. House Representatives from 84th to 117th Congress, or 1955 to 2022, a total of 14,582 Congress-legislator units, and 2,673 distinct legislators. Of them, I focus on 3,860 Congress-legislators (1,546 distinct legislators) who competed in primary elections with at least one other challenger. There were 3,700 victories and 160 defeats.

To measure policy positions, I use roll-call votes, including ideological positions on a liberal-conservative scale and party-line stances. For effort-related behavior, I examine legislative activities that proxy for legislator effort, drawing on previous literature (Dal Bo and Rossi, 2011; Titunik, 2016; Fourinaies and Hall, 2022). These activities include roll-call participation, bill sponsorship and cosponsorship, and the frequency of e-newsletters sent to constituents.

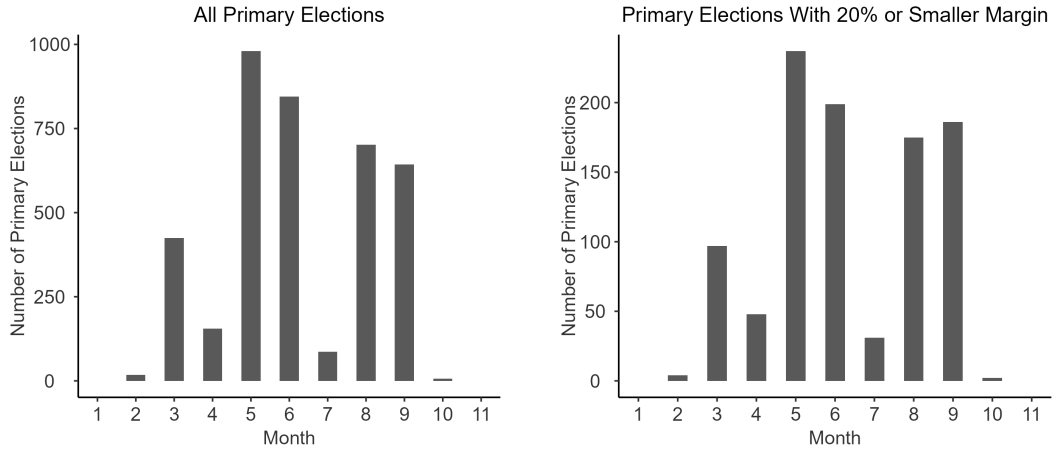
The House representatives' information and roll-call voting records are collected from [voteview.com](http://voteview.com) (Lewis et al., 2024). Sponsorship data combine the Congressional Bills Project's records for 1973 to 2014 (Adler and Wilkerson, 2020) with sponsor records I parse from official Congress.gov bulk data for 2015 to 2022; the Congress.gov records themselves reach back to 2003, and across the six overlapping Congresses the paper draws from the Bills Project (the 108th to 113th), per-legislator counts from the two sources correlate at 0.96, with a median within-legislator ratio of 1.00; Appendix E details the roster restriction and identifier repairs applied to these records. Cosponsorship records combine James Fowler's cosponsorship data for 1979 to 2004 (Fowler, 2006) with records I collect from official Congress.gov bulk data for 2005 to 2022, covering all four types of House bills and resolutions. E-newsletter data come from DCinbox (Cormack, 2024) and cover 2010 to 2022. The designs compare legislators who still face an upcoming general election with legislators who do not, and that contrast is defined only until the general election is held: once it passes, the winner group itself mixes in general-election losers, whose reelection incentives are also extinguished. All outcomes are therefore measured from the start of the Congress through each legislator's own general election date, and the post-general lame-duck weeks are reported separately, as an explicitly labeled extension (Appendix G).<sup>1</sup>

Examining post-primary behavior requires knowing the exact dates of primary elections. The Federal Election Commission provides primary election dates for all states back to 1996,

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<sup>1</sup>Extending the roll-call outcomes over the full Congress, including the post-general lame-duck session, moves the vote-level difference-in-differences participation estimate from 0.105 to 0.115 in absolute value—10.5 to 11.5 percentage points of roll-call participation—both precisely estimated. The appendix opens with a note on the window convention each exhibit uses.

**Figure 1:** Monthly Distribution of Primary Elections



*Note:* The left figure shows the month distribution of all primary elections, while the right figure shows those with less than 20% point margins. In both cases, June was the median month.

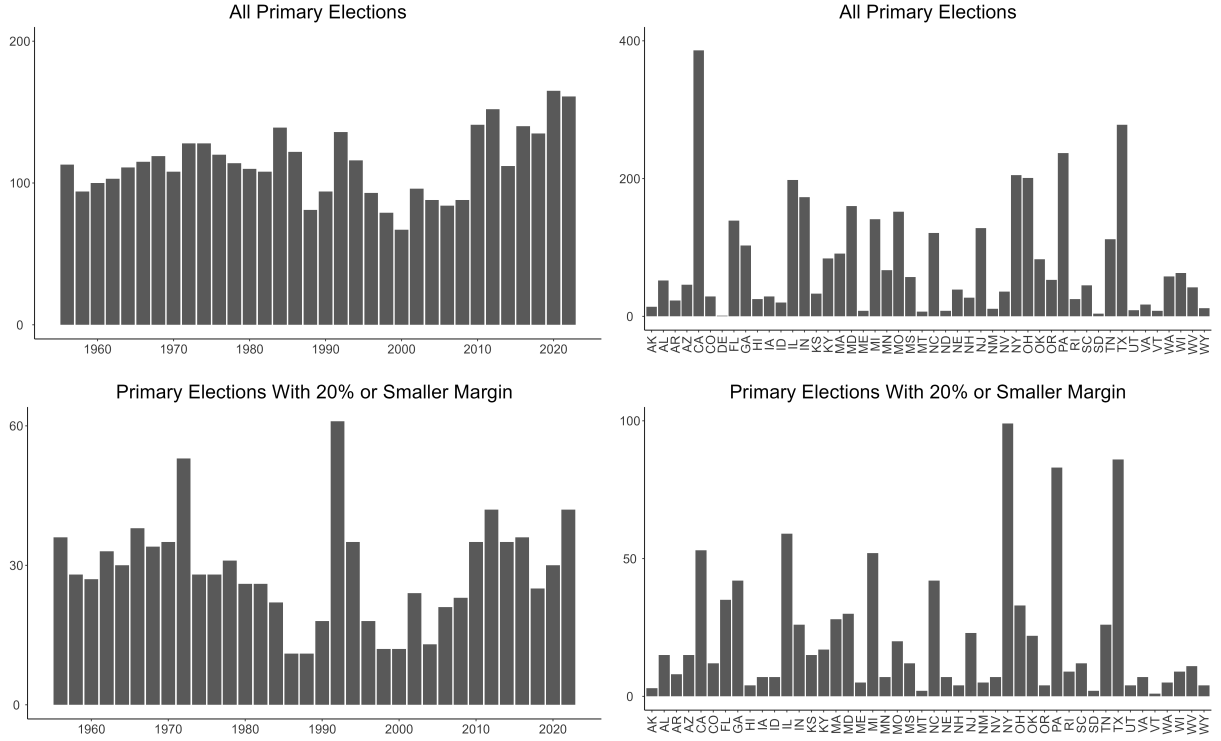
and a few state archives provide earlier dates. For the other years and states back to 1956, I manually collected them from historical newspaper articles.

Figure 1 and Figure 2 show the distribution of primary election dates by month, state, and time. The median month of primary elections was June. A total of about three and a half million roll-call votes were cast by the legislators in the analysis, and 24 percent of those were cast after each legislator’s primary election date.

### 3 Empirical Strategy

To examine the behavior of primary winners and losers, I employ two complementary research designs. A regression discontinuity design compares legislators who narrowly won versus narrowly lost close primary races. A difference-in-differences approach tracks how legislators’ behavior changes across terms in which they won primaries versus terms in which they lost. Both designs aim to ensure comparability between winners and losers: the regression discontinuity design exploits the as-if random outcomes of close primary elections, while the difference-in-differences approach uses within-legislator changes across terms. Each design has its own strengths and weaknesses, and together they provide more robust evidence on

**Figure 2:** Distribution of Primary Elections by Year and State



*Note:* The graphs on the left and right illustrate the number of primary elections by year and by state, respectively. The top rows display the distribution for all primary elections, and the bottom rows show the distribution for those with margins of less than 20%.

the effects of electoral incentives.

### 3.1 Variables and Measurement

The main independent variable distinguishes whether an incumbent House member wins or loses a primary election, capturing the presence or absence of electoral incentives for the general election. The dependent variable tracks how legislators behave after the primary, reflecting their response to the electoral incentives. Specifically, I measure within-term change in behavior from pre-primary to post-primary periods, which discounts varying base levels across legislators and absorbs any pre-primary anticipatory behavior into the baseline, thereby isolating each legislator's reaction to primary outcomes.

This article’s outcome variables fall into two structurally distinct types. Vote-based outcomes—roll-call participation, party-line voting, and ideological positions—are responses to a legislative agenda set externally by party leaders and committee chairs: legislators respond to whichever bills reach the floor. Because the composition of bills scheduled before and after the primary may differ systematically, controlling for bill-level variation is essential. In contrast, non-vote outcomes—bill sponsorship, cosponsorship, and e-newsletters—are activities where legislators can more freely set their own agenda. This structural difference shapes the dependent variable construction and estimation strategy for each type.

To analyze non-vote behaviors, I measure each activity as a weekly rate and compare the rate after the primary with the rate before it. For each legislator-Congress, the pre-primary rate divides the activity count between the start of the Congress and the primary date by the length of that period in weeks, and the post-primary rate does the same for the period between the primary and the general election. The dependent variable is the change in weekly activity,

$$\frac{B_2}{w_2} - \frac{B_1}{w_1}, \tag{1}$$

where  $B_1$  and  $B_2$  are the pre- and post-primary counts and  $w_1$  and  $w_2$  are the lengths of the two windows in weeks. Measuring rates accommodates the fact that primary dates—and hence window lengths—vary across states and years, and anchoring the pre-primary window at the start of the Congress uses each legislator’s full pre-primary record, so that short-lived fluctuations close to the campaign do not dominate the baseline. The window lengths are measured against each activity’s own opportunity set. Introducing and cosponsoring bills require the House to be in session, so for sponsorship and cosponsorship  $w_1$  and  $w_2$  are session-weeks—House session days, from the Congressional Record’s session calendar, divided by five—rather than calendar weeks; a calendar denominator would dilute the rate with weeks in which no bill could be introduced, and the two-month recesses and adjournments concentrate in the post-primary window. E-newsletters go out during recess as well, so their rates remain per calendar week, and roll-call participation is already measured against

the votes actually held. The session calendar, its validation against the House Historian’s legislative days, and calendar-time comparisons for every cell appear in subsection A.19. I apply this measure to legislators’ bill sponsorship, co-sponsorship, and e-newsletter counts in both research designs. The appendix reports a bounded alternative—the proportion of activity occurring after the primary, computed on windows of equal length on both sides of the primary—and discusses where the two measures agree and where they register different margins of the same response; the proportion’s exposure to pre-primary level imbalance is tested directly there (Appendix D, Table D.2).

For roll-call-based behaviors (roll-call participation, voting with the party majority, and voting with party leaders), the two research designs construct outcomes at different units of analysis but estimate the same within-legislator post-primary parameter.

For the regression discontinuity design, I construct a legislator-level dependent variable by estimating, for each Congress  $t$ , the following OLS regression:

$$\text{Vote}_{ipbt} = \gamma_{it} + \delta_{pb} + \beta_{it}\text{AfterPrimary}_{ibt} + \varepsilon_{ipbt}. \quad (2)$$

In this model, the left hand side,  $\text{Vote}_{ipbt}$ , is a binary indicator set to 1 if legislator  $i$  from party  $p$  exhibits a specific voting behavior, such as participating in roll-call voting, on bill  $b$  in Congress  $t$ , and 0 otherwise. The term  $\gamma_{it}$  represents legislator fixed effect and  $\delta_{pb}$  accounts for bill-by-party fixed effects. The key variable,  $\text{AfterPrimary}_{ibt}$ , is a binary indicator that takes the value 1 if legislator  $i$  faces the roll-call for bill  $b$  after the primary election and 0 otherwise. The coefficient  $\beta_{it}$  captures the direction and magnitude of the change in legislator  $i$ ’s voting behavior after the primary in Congress  $t$ , and I use it as the legislator-level outcome in the regression discontinuity specification (Section 3.2.1). I extend this approach to other aspects of roll-call voting behavior by re-estimating the regression with alternative definitions of  $\text{Vote}_{ipbt}$ , such as aligning with the party majority, or voting in line with party leaders.<sup>2</sup>

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<sup>2</sup>When measuring the policy positions based on roll-call votes, I follow existing measurement methods and include roll-call votes where legislators voted either “yea” or “nay,” excluding any abstained votes, so that abstention does not enter the position estimates mechanically. The estimates therefore rest on the votes

For the difference-in-differences design, I construct a bill-vote-level dependent variable by retaining the same bill-level vote data and estimating the post-primary change directly via the interaction between primary outcome and the post-primary indicator (full specification in Section 3.2.3). This avoids the intermediate step of collapsing each legislator-Congress' voting records into a single  $\beta_{it}$  and estimates the same within-legislator parameter; the two approaches yield quantitatively close estimates (Table A.12).

The two designs construct the vote-based dependent variables at different levels of analysis to reflect the structure of their identifying variation. The regression discontinuity design compares different legislators near the threshold, so each legislator-Congress contributes one independent observation; replicating the same running variable across a legislator's bill votes would inflate the sample without adding independent comparisons. The difference-in-differences design instead compares the same legislator across terms, so vote-level observations are natural for the within-legislator comparison and allow bill-by-party-by-Congress fixed effects to absorb bill composition differences.

These approaches for vote-based behavior share an important advantage over a simple pre/post mean comparison: the types of bills introduced before and after primary election dates may differ systematically, so a raw rate comparison could conflate genuine post-primary behavior change with shifts in the legislative agenda. Through bill-party fixed effects within Congress, these dependent variables absorb the bill-by-party level differences, ensuring that the estimated change reflects post-primary lame-duck status itself.

Nevertheless, as a robustness check, I also measure policy positions in the appendix using a simple difference of mean voting patterns between pre- and post-primary periods, with consistent results (Table A.11). The appendix also presents a descriptive comparison of bills before and after the primary. The distribution of bill topics is largely stable across the two periods, with government operations, defense, and public lands remaining the most common categories both before and after the primary (Figure A.3); similarly, weekly vote

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legislators chose to cast; the bill-by-party fixed effects keep the comparison within a given bill, so a thinner post-primary vote set alters the weighting of bills, not the within-bill comparison itself.

volume, participation, and passage rates show no systematic shift around the primary date (Figure A.4). These comparisons thus reveal no notable, systematic differences in the bills considered before and after primary elections.

Roll-call extremism, an additional vote-based outcome, requires one further step beyond this shared construction: each vote is coded by the legislator’s own-party pole (conservative for Republicans, liberal for Democrats), so that higher values indicate own-pole voting. For the regression discontinuity design, I re-estimate equation (2) with  $\text{Vote}_{ipbt}$  set to a conservative-vote indicator—classifying each bill as liberal or conservative using the bill-direction algorithm of Fowler and Hall (2012)—and then sign-flip the resulting  $\beta_{it}$  so that higher values indicate increased conservatism among Republicans or increased liberalism among Democrats. For the difference-in-differences design, the same conservative-vote indicator is sign-flipped for Democrats—yielding the own-party-pole indicator  $\text{extreme}_{iv}$ —so that higher values likewise indicate own-pole voting. Both approaches estimate the same conceptual quantity, i.e., the within-legislator post-primary shift in own-pole voting, and I additionally examine the absolute values of these changes to assess whether legislators adjust positions in either direction even when net shifts cancel.

## 3.2 Research Designs

The research designs exploit the fact that an incumbent must win the party’s primary or be a sole candidate to advance to the general election. While most incumbent House Representatives secure primary election victories with relative ease, this is not always the case. Occasionally, a prominent local figure’s entry into politics poses a threat to the incumbent, and at other times, primary voters view their incumbent politician as old and incompetent, opting to replace them with a new candidate. Changes in district boundaries can also jeopardize an incumbent’s electoral prospects, as they may move loyal supporters to other districts. Consequently, some primary elections result in either the incumbent losing or facing a closely contested race. These variations in outcomes provide the identifying variation for estimating

electoral incentive effects.

### 3.2.1 Regression Discontinuity Design

First, I implement the regression discontinuity design. The running variable  $V_{it}$  is a legislator  $i$ 's share of the top two candidates' vote less 0.5 in the primary election during Congress  $t$ . For example, if an incumbent received 60% of the votes, and two challengers received 35% and 5%, respectively, the running variable is calculated as  $\frac{60\%}{60\%+35\%} - 0.5 = 0.132$ . This indicates that incumbents with positive running variables won primary elections, whereas those with negative running variables lost.

The construction of this running variable is not feasible in states with different primary election systems, so I exclude these states. Specifically, I exclude Georgia before 1962, which used a county-unit system in place of direct popular vote. I also exclude states with nonpartisan blanket primaries, such as the top-two or top-four systems used in Washington (since 2008), California (since 2012), and Alaska (since 2022). Additionally, I exclude Connecticut and Louisiana because their primary systems differ significantly; in Connecticut, party nominees are typically chosen at party conventions, making primaries rarely competitive, while Louisiana uses a nonpartisan blanket primary where winners can be elected outright.

I estimate the standard regression discontinuity specification,

$$Y_{it} = \beta_0 + \beta_1 \mathbf{1}\{V_{it} < 0\} + f(V_{it}) + \varepsilon_{it}. \quad (3)$$

Here,  $V_{it}$  is the running variable, or a legislator's top-two winning margin in the primary election, and  $Y_{it}$  captures the legislator's change in behavior following the primary election. The function  $f(\cdot)$  is flexible in the running variable; given the small sample size, I primarily use a linear specification (Gelman and Imbens, 2019) with a 20-percentage-point bandwidth. In the appendix, I show that the sign and direction of the estimates remain largely stable across alternative fixed bandwidths (Figure A.2), and I report estimates at the data-driven

optimal bandwidth of Calonico, Cattaneo and Titiunik (2014) (Table A.6). Because close primaries are few, those optimal windows are narrow and trade statistical power for local identification—several effort coefficients, including participation, lose significance there—so I treat them as a limitation of the close-primary design rather than as corroboration and retain the 20-percentage-point window for the main analysis. The coefficient of interest,  $\beta_1$ , represents the RD estimator, capturing the effect of primary election *loss*—the removal of electoral incentives that is the substantive comparison of interest—on changes in legislators’ behavior, and it is the coefficient the tables report.

Columns (1) and (2) of Table 2 report means for close losers and close winners, whose primary vote margin was within 20 percentage points; column (5) is the balance test, the estimated jump at the threshold (close losers minus close winners) on the same legislators. No row of column (5) shows a significant jump: neither the pre-determined characteristics nor the vote-based pre-primary behaviors (roll-call participation, voting with the party majority, voting with party leaders) differ significantly at the threshold, and the appendix extends the test to freshman status and the length of the measurement window with the same result (subsection A.1, which opens the appendix with the full balance record). In the window just before the primary, the cosponsorship and e-newsletter means in columns (1) and (2) are higher for close losers, and both estimates at the threshold are positive, but neither is significant at the 10 percent level (+9.4 cosponsored bills, standard error 8.8; +9.1 e-newsletters, 6.8). Over the full pre-primary period, too, no activity differs significantly at the threshold. One Congress earlier the differences that do appear have eventual losers the *less* active group (sponsorship,  $-0.17$  per session-week,  $p = 0.04$ ; e-newsletters,  $-0.40$  per week,  $p = 0.09$ ), while cosponsorship’s estimates, positive there as well, are smaller than their own standard errors on every unit. Both run against, not toward, the pattern that could manufacture a post-primary decline (Table A.2).

Any residual winner–loser difference, even one too small for the balance tests to detect, could in principle enter and bias the estimates through two channels: the legislators them-

**Table 2:** Observable Characteristics of Legislators by Electoral Margin in Primaries

|                                                       | (1)<br>Close<br>Losers | (2)<br>Close<br>Winners | (3)<br>Close<br>Primaries | (4)<br>Non-close<br>Primaries | (5)<br>Loser – Winner<br>at Threshold |
|-------------------------------------------------------|------------------------|-------------------------|---------------------------|-------------------------------|---------------------------------------|
| N                                                     | 159                    | 820                     | 979                       | 2,881                         | 979                                   |
| Congress                                              | 100.08 (10.31)         | 100.16 (10.39)          | 100.14 (10.37)            | 101.34 (10.04)                | −0.12 (1.58)                          |
| Seniority                                             | 5.11 (3.31)            | 5.04 (4.01)             | 5.05 (3.90)               | 5.06 (3.90)                   | 0.13 (0.60)                           |
| Age                                                   | 57.99 (10.98)          | 56.27 (10.97)           | 56.55 (10.98)             | 55.40 (11.01)                 | 1.18 (1.68)                           |
| Democratic party                                      | 0.65                   | 0.62                    | 0.63                      | 0.62                          | −0.000 (0.074)                        |
| Previous vote share in general                        | 0.68 (0.15)            | 0.71 (0.16)             | 0.70 (0.15)               | 0.70 (0.15)                   | −0.004 (0.024)                        |
| DW-NOMINATE score                                     | −0.08 (0.40)           | −0.05 (0.39)            | −0.05 (0.40)              | −0.06 (0.40)                  | 0.023 (0.061)                         |
| DW-NOMINATE $\times (-\mathbf{1}\{\text{Democrat}\})$ | 0.37 (0.18)            | 0.35 (0.19)             | 0.35 (0.19)               | 0.37 (0.17)                   | −0.025 (0.029)                        |
| Roll-call participation before primary                | 0.89 (0.11)            | 0.91 (0.08)             | 0.91 (0.09)               | 0.94 (0.07)                   | −0.018 (0.013)                        |
| Sponsored bills pre-primary                           | 3.24 (4.04)            | 3.07 (4.15)             | 3.09 (4.13)               | 3.09 (3.95)                   | 0.23 (0.80)                           |
| Cosponsored bills before primary                      | 66.03 (46.83)          | 57.07 (41.24)           | 58.52 (42.29)             | 57.89 (45.30)                 | 9.42 (8.80)                           |
| E-newsletters before primary                          | 21.40 (38.36)          | 14.88 (16.74)           | 15.95 (21.80)             | 11.73 (14.67)                 | 9.08 (6.80)                           |
| Voting with party majority before primary             | 0.87 (0.10)            | 0.87 (0.10)             | 0.87 (0.10)               | 0.89 (0.09)                   | −0.016 (0.016)                        |
| Voting with party leaders before primary              | 0.88 (0.10)            | 0.88 (0.11)             | 0.88 (0.11)               | 0.90 (0.09)                   | −0.020 (0.017)                        |

*Note:* Close primaries are those with margins smaller than 20 percentage points. Standard deviations are in the parentheses in columns (1) to (4). Column (5) reports the local-linear regression discontinuity estimate at the threshold (close losers minus close winners, 20-percentage-point bandwidth, separate slopes on each side), with its standard error in parentheses, estimated on the same legislators as columns (1) and (2); because it compares the two groups at a margin of zero rather than across the whole window, it is not the difference between the column (1) and (2) means. Seniority counts Congresses served including the current one; Table A.1 counts prior Congresses only and drops the 84th Congress, where prior service is not observed, so its seniority estimate uses 943 observations. Count activities are measured over the equidistant pre-primary window and cover each activity’s record years (sponsorship 1973–2022, cosponsorship 1979–2022, e-newsletters 2010–2022, with non-users counted as zeros), so their cells rest on the correspondingly smaller subsamples.

selves may differ (eventual losers simply a different kind of representative throughout), or their activity may already be moving within the loss term as the primary approaches. The first channel is neutralized by construction and tested outside the loss term. By construction, a legislator’s baseline level does not enter the estimates by itself: the RDD is estimated on the change in each legislator’s own weekly rate, and the DiD compares each legislator’s post-primary weeks with the same legislator’s pre-primary weeks in that term; what enters is the change against that baseline. The test is whether the upcoming primary’s outcome

explains behavior one Congress earlier, where the loss it assigns cannot yet operate. For levels, the preceding balance record already answers: the only previous-Congress cells away from zero run opposite, eventual losers the less active group. For patterns, recomputing every change estimate around each legislator’s own previous-Congress primary—the same pre/post contrast, one term removed—returns null in all fourteen cells, the one marginal cell opposite in sign (Table A.21).

The second channel is examined inside the loss term, where the event studies of Section 4 confront it directly: the pre-primary leads are flat for participation and sponsorship. The remaining two outcomes, cosponsorship and e-newsletters, depart from this pattern. Cosponsorship’s leads fail flatness because eventual losers begin scaling back in the final pre-primary months while winners’ own path stays flat, a movement that breaks discretely at each legislator’s own primary date and carries losers below their own earlier baseline, the trajectory of a response taking hold rather than of a prior difference reverting (Appendix C); e-newsletters diverge modestly in the final campaign month (Appendix B). Each departure bears on the interpretation of that outcome’s own result; Section 4 therefore weighs them where those results are presented, so that estimate and caveat can be read together. Finally, both channels are tested by the same direct adjustment: each change coefficient is re-estimated with the legislator’s own pre-primary level added as a control. To the extent a coefficient reflected the first channel (a stable difference between the legislators themselves), the level measured over the entire pre-primary period would absorb it; to the extent it reflected the second (movement already underway), the level measured over the outcome’s own window ending at the primary would. None moves materially under either control (subsection A.3).

### **3.2.2 External Validity of Regression Discontinuity Design**

External validity warrants further consideration. The regression discontinuity design results are derived exclusively from legislators who faced close primary elections, which may limit their generalizability to the broader population of legislative representatives. To address this

concern, I also estimate a difference-in-differences specification (Section 3.2.3) that uses all contested primaries, not only close ones, and produces consistent results. As a more direct check on scope, I compare the contested-primary incumbents who enter the design against the non-challenged incumbents who fall outside it—the group that most concretely stands in for that broader population (subsection A.15). The two groups differ only modestly on observable characteristics: the two district-political variables of greatest concern, general-election margin and district presidential vote, differ by just 1.1 and 2.6 percentage points. More tellingly, in a three-group difference-in-differences comparison the non-challenged incumbents behave like narrow primary winners rather than losers across twelve effort and ideology outcomes—exactly what the lame-duck mechanism predicts for legislators who never carried a re-election threat to remove, and an indication that the post-primary effort decline is concentrated specifically among legislators whose re-election incentive disappears. The following considerations further suggest that findings from close primary elections may reasonably extrapolate to legislators in general, including those facing minimal or no primary challenges.

First, representatives in close and non-close primary elections appear largely similar on descriptive characteristics. Columns (3) and (4) of Table 2 present legislators’ characteristics and behavior before the primary elections, distinguishing between those from close and non-close primary contests. Overall, the two groups appear similar across most observable variables, with e-newsletter counts showing the only statistically significant difference between the two groups.

Second, the motivations for primary challenges are similar across close and non-close contests. Using Cowburn (2024)’s classification of challenge motivations for U.S. House Representatives from 2006 to 2020 (following Boatright, 2014), I find that the underlying reasons are largely similar across the two groups. Among close primary contests within 20% point margin, the most common reason for a challenge is the perception that the incumbent is too moderate (56%), followed by concerns over age and competence (17%), the incumbent

being seen as too extreme (13%), scandals (5%), and national issues (4%). This distribution is not dramatically different from that of non-close primaries, where the most frequently cited reason remains the incumbent’s moderate ideology (45%), followed by age and competence (24%), the incumbent being perceived as too extreme (14%), national issues (4%), and scandals (under 1%).

Third, many close primary elections occur as a consequence of redistricting. Redistricting often disrupts an incumbent’s electoral advantages unexpectedly by forcing incumbents to run against one another or by splitting their loyal voter bases across districts. Because these disruptions often lie beyond the influence of legislators’ own strategic decisions or qualities, close primary contests in these instances are less likely to reflect systematic differences between legislators alone. The data confirm this pattern: on average, about 40 House representatives face a primary election decided within a 20% point margin during a redistricting year, compared to about 26 in a non-redistricting year. To the extent these challenges arise from unforeseen external shocks, the dynamics observed in close elections can reasonably generalize to legislators facing little or no primary competition; in a robustness check, I restrict the analyses to redistricting years only, with consistent results (Table A.7).

### **3.2.3 Difference-in-Differences Design**

To demonstrate that the results are not a product of the small sample of close primary elections, I also employ a difference-in-differences design. In this research design, I compare individual legislators’ behavior across terms when they won primary elections versus terms when they lost. The advantage of this design is its ability to utilize all contested primaries rather than only close ones—the same 3,860 contested legislator-terms described in Section 2, that is, incumbents who competed against at least one challenger, now across the full range of primary margins—thereby representing a larger number of legislators and offering greater statistical power. As shown in the results, both designs yield similar conclusions, reinforcing the findings and providing robust evidence. The wide-margin pool does not drive the agree-

ment between the designs: excluding near-uncontested landslide winners, those with winning margins above 45 percentage points, leaves the vote-level participation estimate unchanged at 0.105 in absolute value—about ten percentage points of roll-call participation—and restricting the panel to close races decided within twenty points yields 0.103.

The difference-in-differences specification differs across outcome types, reflecting the dependent-variable structure introduced in Section 3.1. For roll-call-based outcomes (roll-call participation, voting with the party majority, voting with party leaders, and roll-call extremism), I estimate legislators’ change in behavior after primary at the legislator-bill-vote level. Let  $Y_{ibt}$  denote legislator  $i$ ’s voting outcome on roll-call vote  $b$  in Congress  $t$ ,  $\text{PrimaryLoss}_{it}$  an indicator for whether the legislator lost the primary in the Congress  $t$  covering vote  $b$ , and  $\text{AfterPrimary}_{ibt}$  an indicator for whether the vote occurred after that primary’s election date. I estimate

$$Y_{ibt} = \beta_0 + \beta_1 \text{PrimaryLoss}_{it} + \beta_2 \text{AfterPrimary}_{ibt} + \beta_3 (\text{PrimaryLoss}_{it} \times \text{AfterPrimary}_{ibt}) + \delta_i + \gamma_{bpt} + \varepsilon_{ibt}, \quad (4)$$

where  $\delta_i$  denotes legislator fixed effects and  $\gamma_{bpt}$  denotes bill-by-party-by-Congress fixed effects (with  $p$  the legislator’s party), which absorb bill-level and party-specific shocks. Standard errors are clustered at the legislator level. The coefficient of interest is the interaction  $\beta_3$ , which identifies the difference in the post-primary change in voting behavior between terms in which a legislator lost versus won the primary, net of time-invariant differences across legislators.

For non-vote outcomes (sponsored bills, cosponsored bills, and e-newsletters)—where legislators can more freely set their own agenda, as discussed in Section 3.1—the vote-level specification in equation (4) is not appropriate: these activities are each legislator’s own output rather than responses to a common set of bills, so there is no shared bill-level structure on which to define the  $\gamma_{bpt}$  fixed effects. For these outcomes I instead estimate a

weekly count panel. Each legislator-Congress is expanded into weekly activity counts from the start of the Congress through the general election, on a common grid of Wednesday-to-Tuesday weeks anchored at each Congress’s first Wednesday—general elections fall on Tuesdays, so the term ends on a week boundary—and, for sponsorship and cosponsorship, restricted to the weeks with at least one House session day, per the session-time convention of Section 3.1. I estimate

$$Y_{iwt} = \beta_0 + \beta_2 \text{AfterPrimary}_{iwt} + \beta_3 (\text{PrimaryLoss}_{it} \times \text{AfterPrimary}_{iwt}) + \delta_{it} + \gamma_{pwt} + \varepsilon_{iwt}, \quad (5)$$

where  $Y_{iwt}$  is legislator  $i$ ’s activity count in week  $w$  of Congress  $t$ ,  $\delta_{it}$  are legislator-Congress fixed effects, and  $\gamma_{pwt}$  are party-by-Congress-by-week fixed effects—the count analog of the bill-by-party-by-Congress fixed effects in equation (4). Because primary dates are staggered across states, legislators of the same party are observed in the same calendar week both before and after their own primaries, so  $\beta_3$  compares winners’ and losers’ within-term changes in weekly activity while absorbing party-specific agenda and calendar shocks.  $\text{AfterPrimary}_{iwt}$  equals one for the weeks that begin after the primary date. House primaries are nearly all held on Tuesdays—the last day of the week on this grid—so for over ninety percent of the sample the pre/post boundary falls exactly on the primary date; in the remaining states, whose primaries fall on other weekdays, the week containing the primary is coded pre-primary, and dropping those weeks leaves every estimate essentially unchanged. Standard errors are clustered by legislator; coarser time fixed effects yield the same conclusions. Because legislators differ widely in baseline activity volume and the weekly counts are sparse, equation (5) is also estimated by fixed-effects Poisson regression, which compares each legislator’s post-primary weeks against that legislator’s own baseline rate—treating the pre-primary period as data rather than as an ingredient of the outcome—and measures the change in proportional terms; these estimates are reported alongside the least-squares estimates for the count outcomes.<sup>3</sup>

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<sup>3</sup>Fixed-effects Poisson estimation uses only observations that carry information about a proportional change, which is why the Poisson columns report smaller sample sizes than their least-squares twins. For sponsorship and cosponsorship the difference consists mostly of weeks in which no legislator of a party records

For the vote-based outcomes, an aggregated legislator-level specification is also reported in the printed appendix (Table A.12) as a cross-check.

## 4 Primary Losers Exhibit Decreased Effort

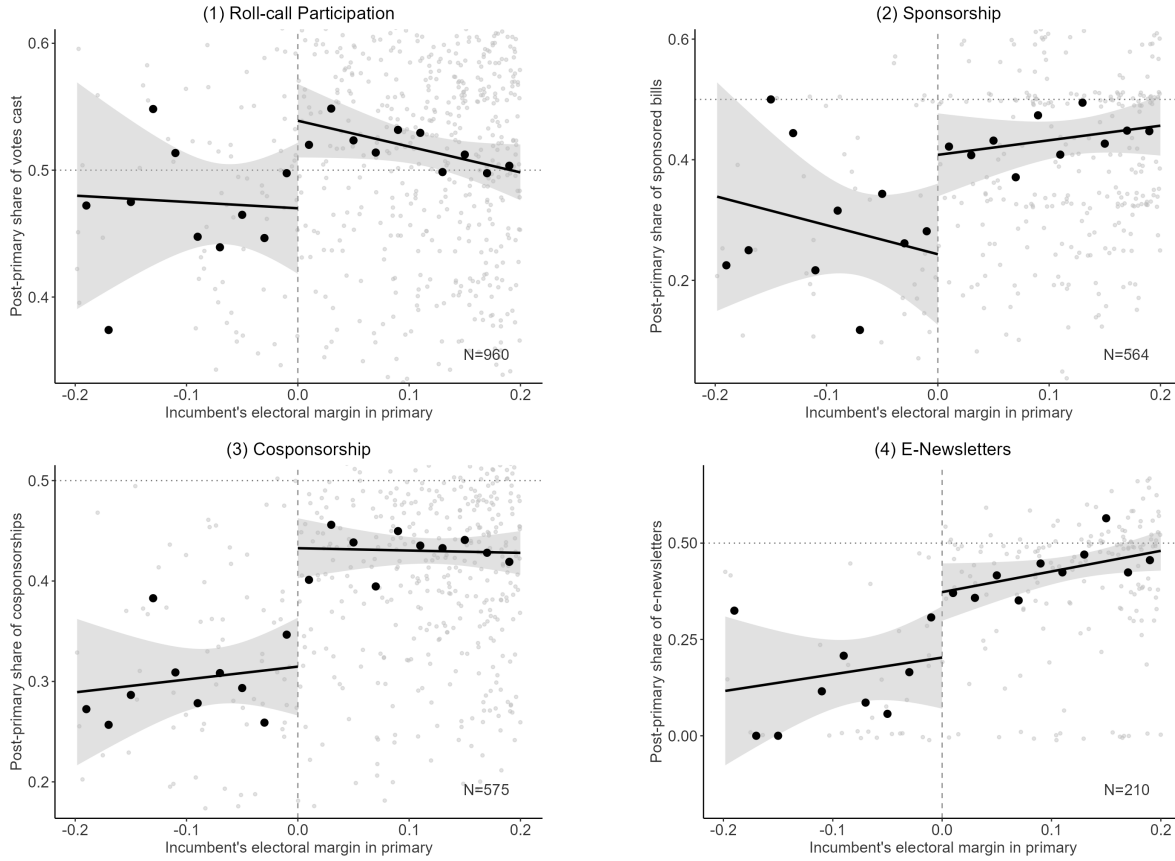
Figure 3 presents the discontinuity for all four effort activities, each measured as the share of a legislator’s activity that falls after the primary: all four drop for legislators who fall just short of renomination, with a clear jump at the threshold between barely losing and barely winning. Measured in absolute terms rather than against a legislator’s own volume, sponsorship alone shows no discontinuity (Figure D.1). Figure 4 shows when the gaps emerge. For participation, winners and losers are indistinguishable for the first fourteen months of the term; the gap opens in the final weeks of the primary campaign, widens sharply at the primary itself, and persists through the general election; in the labeled full-term extension the gap remains—indeed grows slightly—through the post-general lame-duck weeks (Appendix G). The count outcomes break at the primary as well, each with the qualification its own paragraph below discusses.

This timing pattern is the signature of an incentive effect rather than a career winding down. Because primary dates are staggered across states, the event-study panel nets out calendar-time shocks; a gradual winding down of careers would produce a slope spread over the whole term, not a flat year followed by a break aligned to each legislator’s own primary date. The modest pre-primary onset reflects the campaign itself rather than anticipatory disengagement by eventual losers: closely contested incumbents, winners and losers alike, spend the final weeks campaigning at home relative to safe winners, and the participation

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any activity—overwhelmingly weeks the House is in recess—which enter a least-squares average as zeros but are absorbed as uninformative by the Poisson week fixed effects. The remainder, and for e-newsletters the larger part, are legislator-Congress cells with zero activity in every week (27 such cells for sponsorship, 7 for cosponsorship, 95 for e-newsletters). That drop is bookkeeping rather than sample selection: for a legislator who records no activity at all in a Congress, a proportional change is not defined—there is no rate whose movement the estimator could measure—so these cells are naturally excluded. Legislators active before the primary and silent after it, the sharpest form of withdrawal, have a well-defined proportional change and remain in the sample; only the never-active drop out.

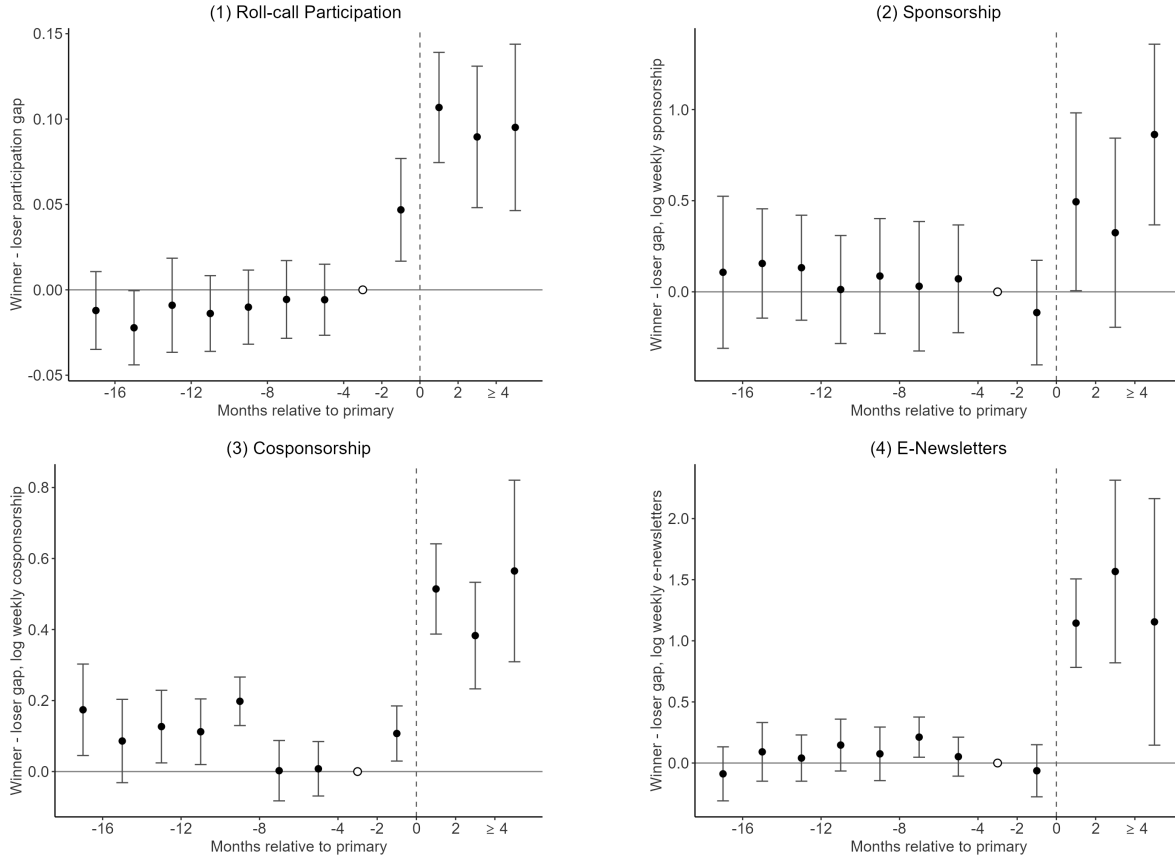
**Figure 3:** Regression Discontinuity Results on Legislators' Effort-Related Activities: Proportional Margin



*Note:* Each panel plots the share of the activity that falls after the primary,  $\text{post}/(\text{pre} + \text{post})$ , against the incumbent's primary winning margin; legislators to the left of the threshold lose the primary. The windows are equidistant around the primary, and the dotted line at 0.5 marks activity unchanged from before it. Large black points are averages within 0.02-point bins, small gray points the raw data jittered vertically for legibility, and the lines are OLS fits estimated separately on each side; each panel's display range is set around the binned means and fitted lines so the estimated discontinuity is legible, and raw points outside that range are not shown (fits use all observations). Sample sizes match the proportional-margin cells of Table 3; the additive-margin panels, on which sponsorship is null, appear in Figure D.1. The share is undefined for a legislator with no activity in either window, which drops one legislator-term for participation, 101 for sponsorship, 3 for cosponsorship, and 35 for e-newsletters—the least active members in each case. Sponsorship's share takes few distinct values, since the median legislator introduces about 0.16 bills per week, so its raw points fall on discrete levels.

dip appears on both sides of the eventual result—a pattern anticipation cannot produce, since eventual winners have no defeat to foresee. Consistent with this reading, the onset shrinks toward zero as the margin window tightens and is absent among primaries decided within

**Figure 4:** Timing of the Effort Response: Event Studies around the Primary



*Note:* Each panel traces the winner–loser gap in the activity by event time relative to each legislator’s own primary date, in two-month bins with 95% confidence intervals; the open circle marks the omitted reference bin (months  $-4$  to  $-2$ ), estimates are plotted at window centers so the dashed line at the primary date separates pre from post, the edge bins pool months  $-16$  and earlier and  $+4$  and later, and the post window ends at each legislator’s general election. Panel (1) is the close-primary participation event study on the additive margin, participation’s claim margin (20-percentage-point window; full-term extension in Appendix G). Panels (2)–(4) are the proportional-margin count event studies—fixed-effects Poisson on the weekly count panels (log scale), event-time and party-by-Congress-by-week fixed effects, all contested primaries (sponsorship Congresses 93–117, cosponsorship 96–117, e-newsletters 2010–2022); the week fixed effect absorbs each week’s session exposure, and the additive-margin panels appear in Figure D.2. Standard errors are clustered by legislator. Joint tests that the pre-reference leads are flat:  $p = 0.46$  (participation),  $0.97$  (sponsorship),  $< 0.001$  (cosponsorship),  $0.004$  (e-newsletters); joint post-primary gaps:  $p < 0.01$  for all four, and sponsorship’s proportional gap is deepest in the pooled final bin. Cosponsorship’s pre-primary movement comes entirely from the eventual losers’ side—winners’ own path is flat; the group trajectories are decomposed in Appendix C. The tabulated event studies behind the pre-registered timing criteria appear in Table B.1 and Table C.2, with the four-week pre-registered facts preserved in their notes.

ten points, races in which both sides campaign comparably hard, while the discontinuity at the primary itself persists in every window (Appendix G). A placebo exercise reinforces the

point: displacing the comparison window to arbitrary dates away from the primary produces no effect, while the actual primary date does (subsection A.20).

Table 3 presents the effort estimates under both research designs, expressed as the effect of losing the primary. For each activity, both designs are estimated on both margins of measurement: the additive margin, the absolute change in activity, and the proportional margin, the change relative to each legislator’s own volume. The two margins have power in different places, so the table reports every combination rather than privileging one, and the paper claims what survives every informative measurement: a margin too coarse to detect the change at issue can withhold corroboration, but its null is not read as contrary evidence. For roll-call participation the two margins are two readings of the same quantity rather than independent checks—the chamber sets the denominator, the votes held, common to every legislator in a week—so its proportional estimates restate the additive ones against the base rate and are reported for completeness. Every estimate in the table carries a previous-Congress placebo, re-estimated one Congress earlier where no incentive effect exists to find; the placebos, reported in full in subsection A.21, are null throughout, and the single marginally significant cell runs opposite in sign to the estimates it checks. The two designs, resting on different comparisons and different samples, agree closely wherever both carry information. Decomposing the discontinuity confirms the pattern the design predicts: at the cutoff, winners and losers are indistinguishable in their pre-primary participation and cosponsorship levels, and the entire gap arises in the post-primary level (Table A.3).

Legislators who lose their primary election become approximately 10 to 12 percentage points less likely to participate in roll-call votes. E-newsletter communication with constituents falls by roughly two-thirds of a newsletter per week—nearly three newsletters a month—in the same post-primary pattern. The e-newsletter finding rests on the shortest record: the data cover only 2010–2022, the close-primary sample is correspondingly small, and winners’ and losers’ e-newsletter activity also diverges modestly in the final campaign month before the primary (Appendix B). A variety of placebo checks confirm the head-

**Table 3:** Primary Loss and Legislators’ Effort-Related Activities: Two Designs, Two Margins

| Regression Discontinuity                                          |                                                     |                                                    |                                                      |                                           |
|-------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|------------------------------------------------------|-------------------------------------------|
|                                                                   | Roll-call<br>Participation<br>(share of votes cast) | Bill<br>Sponsorship<br>(count per<br>session-week) | Bill<br>Cosponsorship<br>(count per<br>session-week) | E-Newsletters<br>(count per week)         |
| <i>Additive margin: change in rate</i>                            |                                                     |                                                    |                                                      |                                           |
| Primary Loss                                                      | −0.119***<br>(0.020)                                | −0.024<br>(0.075)                                  | −1.521***<br>(0.537)                                 | −0.639***<br>(0.198)                      |
| Observations                                                      | 926                                                 | 665                                                | 578                                                  | 245                                       |
| <i>Proportional margin: post-primary share of activity</i>        |                                                     |                                                    |                                                      |                                           |
| Primary Loss                                                      | −0.069**<br>(0.028)                                 | −0.165**<br>(0.068)                                | −0.118***<br>(0.030)                                 | −0.170**<br>(0.072)                       |
| Observations                                                      | 960                                                 | 564                                                | 575                                                  | 210                                       |
| Bandwidth                                                         | 0.2                                                 | 0.2                                                | 0.2                                                  | 0.2                                       |
| Kernel                                                            | Uniform                                             | Uniform                                            | Uniform                                              | Uniform                                   |
| Specification                                                     | Linear                                              | Linear                                             | Linear                                               | Linear                                    |
| Difference in Differences                                         |                                                     |                                                    |                                                      |                                           |
|                                                                   | Roll-call<br>Participation<br>(share of votes cast) | Bill<br>Sponsorship<br>(count per<br>session-week) | Bill<br>Cosponsorship<br>(count per<br>session-week) | E-Newsletters<br>(count per week)         |
| <i>Additive margin: change in rate (least squares)</i>            |                                                     |                                                    |                                                      |                                           |
| Primary Loss × Post                                               | −0.105***<br>(0.014)                                | −0.049**<br>(0.022)                                | −1.026***<br>(0.173)                                 | −0.673***<br>(0.204)                      |
| Observations                                                      | 3,520,416                                           | 236,592                                            | 206,296                                              | 96,158                                    |
| <i>Proportional margin: change relative to own rate (Poisson)</i> |                                                     |                                                    |                                                      |                                           |
| Primary Loss × Post                                               | −0.125***<br>(0.018)                                | −0.441**<br>(0.172)                                | −0.387***<br>(0.061)                                 | −1.223***<br>(0.241)                      |
| Observations                                                      | 3,520,416                                           | 226,627                                            | 209,231                                              | 81,840                                    |
| Unit of Analysis                                                  | Legislator-vote                                     | Legislator-week                                    | Legislator-week                                      | Legislator-week                           |
| Fixed Effects                                                     | Legislator,<br>Bill-Party-<br>Cong.                 | Legislator-Cong.,<br>Party-Cong.-<br>Week          | Legislator-Cong.,<br>Party-Cong.-<br>Week            | Legislator-Cong.,<br>Party-Cong.-<br>Week |

\* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ 

*Note:* Coefficients are the effect of losing the primary (Primary Loss =  $\mathbf{1}\{V_{it} < 0\}$ ); standard errors in parentheses; each cell is a separate estimation. Additive-margin estimates measure absolute changes (equations (3) and (1) at the cutoff; the vote-level interaction of equation (4) and least squares on the weekly count panel of equation (5)). For sponsorship and cosponsorship the additive cells are measured over session time—cutoff rates per session-week of five House session days, and panel weeks restricted to those with at least one session day—per the measurement convention of Section 2; e-newsletters, which do not require the chamber, are per calendar week, and calendar-time estimates for all cells appear in subsection A.19.

Proportional-margin estimates measure change relative to each legislator’s own volume: the post-primary proportion post/(pre + post) at the cutoff (votes cast, for participation; windows equidistant around the primary, with 0.5 the benchmark of unchanged activity), and the same panel re-estimated by fixed-effects Poisson—the weekly count panel for the count outcomes, the vote-level panel of equation (4) for participation (log scale, with  $\exp(b) - 1$  the proportional change in a legislator’s own rate; no-activity weeks and all-zero cells drop out as bookkeeping, hence the smaller count Poisson samples).

Table D.2 holds the windows fixed across margins and reports the proportion’s randomization inference. For participation the two margins are mechanically linked rather than independent—the denominator, the votes held, is common to every legislator in a week—so its proportional cells are the additive estimates translated through the base rate (the Poisson coefficient of  $-0.125$  translates to an 11.8 percent decline in a legislator’s own rate; the least-squares estimate of  $-0.105$ , set against the pre-primary rate it implies—the losers’ post-primary mean of 0.79 plus the decline itself—is the same 11.7 percent); they are shown for completeness of the margin map, and their agreement is arithmetic rather than independent corroboration. Winner and loser mean changes at the cutoff are 0.029 and  $-0.086$  for participation, 0.076 and  $-0.611$  for e-newsletters. The previous-Congress placebos are reported in full in Table A.20 (fourteen cells; the vote-level difference-in-differences placebos are not constructed), thirteen of fourteen null and the one marginal cell opposite in sign to the estimates.

Regression-discontinuity standard errors are conventional (bootstrap alternatives in Table A.22); panel standard errors are clustered by legislator. Sponsorship records cover 1973–2022, cosponsorship 1979–2022, e-newsletters 2010–2022. The weekly panel remains the primary specification for the count conclusions; sponsorship’s decline is claimed on the proportional margin, the session-time panel detecting it on the additive margin as well, and its cutoff nulls are the arithmetic consequence of its low

line finding, most decisively for participation, including a placebo-window test showing that the effort effects are specific to the actual primary date rather than to an arbitrary short window, and a previous-Congress placebo showing that the same change is absent one term earlier (subsection A.2, subsection A.20, subsection A.21); the count results are additionally invariant to the window construction (Appendix D).

These findings closely align with conventional literature that argues legislators’ effort-related behavior is driven by their reelection motivation (Mayhew, 2004; Fenno, 1978; Cox and McCubbins, 2007). Conversely, it contrasts with predictions by scholars who contend that voters’ insufficient attention and political knowledge constrain representatives’ responsiveness and thus the effects of electoral incentives are limited (Achen and Bartels, 2017; Somin, 2016).

Nevertheless, lame duck legislators continue to participate in most roll-call votes and engage in various legislative activities, even after losing their primaries, albeit at a lower frequency than winners. After losing their primaries and until the general election, on average, these legislators still participate in 79% of roll-call votes, sponsor 1.66 bills, cosponsor 44.83 bills, and send 6.94 e-newsletters. These figures indicate that electoral incentives are hardly the sole driver of legislative activities. Their continued engagement may reflect ideological commitment, policy advocacy, reputation concerns for post-Congress careers, or a combination of these and other motivations.

Bill sponsorship, the most consequential of the count activities and the sparsest, is the outcome where measurement decides what is visible (Table 3). Measured over calendar weeks, its absolute response is barely visible: the weekly-rate discontinuity is null, and the calendar-week panel estimate reaches only the ten-percent level. Part of that faintness is arithmetic—the median legislator introduces about 0.16 bills per week, so even a decline as deep as a third of a legislator’s own rate implies an absolute change of a few hundredths of a bill—and part of it is the calendar: weeks in which the House holds no session enter a calendar-week average as mechanical zeros for winners and losers alike, diluting the gap. Measured over the weeks

the House sits—the session-time convention of Section 2—the panel detects the absolute decline ( $-0.049$  bills per session-week,  $p = 0.027$ ) with a clean previous-Congress placebo, while the cutoff discontinuity remains null, lacking the panel’s power. Measured against each legislator’s own baseline, the decline is detected in both designs: the fixed-effects Poisson estimate of equation (5) is  $-0.44$  ( $p = 0.010$ ;  $-0.40$  among close races), and the bounded proportion falls at the cutoff ( $-0.165$ ,  $p = 0.016$ ). Both carry null previous-Congress placebos and are sharp outliers under displaced-date randomization inference, and the cutoff estimate is unchanged by adjustment for the pre-primary level (Appendix D).<sup>4</sup> The timing agrees across margins once the measurement does: the session-week event study’s post-primary gaps are jointly significant at the ten-percent level ( $p = 0.056$ ) with flat leads, and on a base that itself shrinks toward the session’s end, the proportional gap if anything deepens late in the term (Appendix C). Nor is the proportional response a ceremonial artifact. A decomposition specified in advance, using a title classifier validated against the Center for Effective Lawmaking’s commemorative counts (Volden and Wiseman, 2014), locates the decline in substantive bills: the substantive-only Poisson estimate is  $-0.44$  ( $p = 0.016$ ) and the substantive-only proportion at the cutoff agrees in direction, while commemorative bills, seven percent of the records, are too sparse to be informative on any margin (Table C.3). Sponsorship, then, declines as the other three activities do, by between roughly a third and a half of a legislator’s own rate: the Poisson estimate of  $-0.44$  translates to a decline of 36 percent ( $\exp(-0.44) - 1$ ), and the cutoff estimate places the post-primary share of activity at 0.335 against the unchanged benchmark of 0.5, a post-to-pre ratio of about one half. The claim is stated on the proportional margin, where every powered design detects it; the session-time panel corroborates it in absolute terms, and the remaining nulls—the cutoff cells—are the arithmetic consequence of the base rate rather than evidence against the

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<sup>4</sup>A cross-sectional Poisson discontinuity—the proportional analog of the level RDD—cannot arbitrate between the margins: comparing across legislators, its minimum detectable effect is roughly twice the size of the panel estimate, and even cosponsorship, whose panel estimate is the strongest of the three count activities, returns a null under it. Its silence for sponsorship is non-corroboration by an uninformative design, not contradiction.

response. The proportional characterization emerged from mapping the outcome’s sensitivity to measurement rather than from criteria specified in advance; it is claimed on the validation reported above.

Cosponsorship, with records covering 1979–2022 and measured as session-week rates, also declines when the reelection incentive is removed, on both margins and in both designs (Table 3). Losing a primary is associated with significantly less cosponsoring at the discontinuity, no such gap appears one Congress earlier, and the panel estimate holds among close races. At the narrow data-driven bandwidth the discontinuity itself is too imprecise to be informative, as for all count outcomes (Table A.6); the close-race panel estimate and the placebo-window test do not depend on the window choice (subsection A.20). The timing evidence carries one qualification that participation’s does not: the event-study leads are not jointly flat, so cosponsorship fails the timing criterion I specified in advance, and the timing decomposition that follows is accordingly post-hoc. That decomposition attributes the lead violation entirely to the losers’ own path. Winners’ cosponsoring is flat throughout the term. Eventual losers rise above their final pre-primary pace over the contest season, peak three to four months out, and converge back to that pace as the primary nears; at the primary they drop discretely, to nearly one cosponsorship per week below it, and stay roughly a third below their own rate through the general election ( $p = 0.004$ ; Appendix C). Read as mean reversion, that path has the wrong destination: reversion returns unusually active legislators to their own normal pace, and losers end the term well below theirs. A prior-year comparison fixes that normal from outside the contest season: measured against the same calendar window one year earlier, close losers fall to well under half of their own prior-year cosponsoring volume while winners hold near two-thirds of theirs—a gap that is precisely estimated ( $p = 0.004$ ) and absent one Congress earlier (Table D.3)—and the retained proportion estimate is essentially unchanged when the pre-primary level is controlled (Appendix D). The trajectory, then, reads as a withdrawal that begins as the loss becomes foreseeable and breaks discretely at the primary itself, not as a return to form. That antici-

patory reading is specific to cosponsorship, and it is consistent with the campaign account of the participation onset: the participation dip appears among closely contested winners and losers alike because campaigning removes members from the floor votes they must attend in person (Appendix G), whereas winners’ flat cosponsoring path through the same campaign weeks shows that position-taking is not mechanically interrupted by campaigning—so a pre-primary decline in cosponsorship isolates the legislators with reason to stop investing in it, those who foresee defeat.

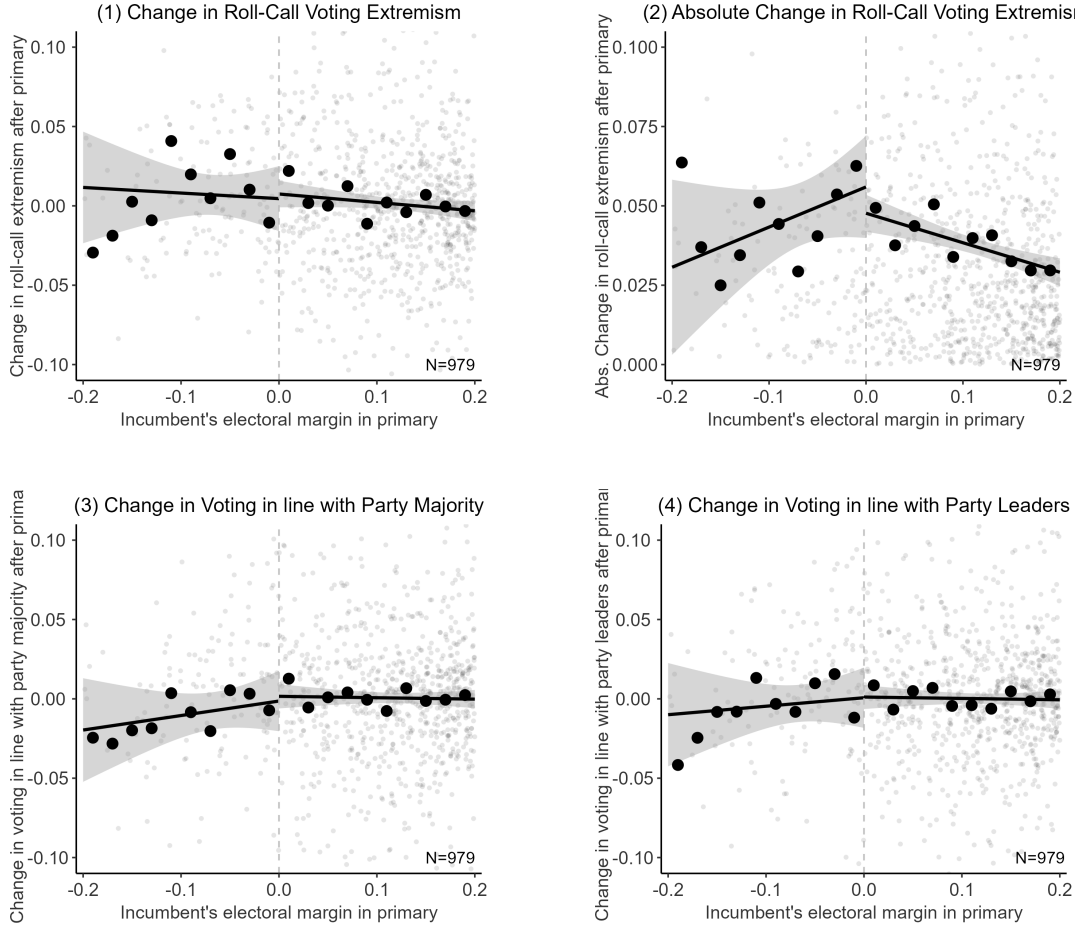
The durable contrast between the two bill-based activities is corroboration and anticipation, not raw magnitude: on the common Poisson translation the two declines are alike, roughly a third of a legislator’s own rate. Cosponsoring—cheap, discretionary position-taking—falls on both margins at the cutoff and in the panel, begins its decline as the loss becomes foreseeable, and stays down through the general election. Sponsorship runs through a longer pipeline of drafting and agenda-setting and is sparse enough that only the panel has the power to see its absolute decline; where a design has power, the response appears, with flat leads and a break at the primary itself. Electoral incentives, on this evidence, take hold earlier in the activity legislators can adjust at will.

## 5 Both Winners and Losers Show No Significant Shift in Policy Positions

Figure 5 visualizes the regression discontinuity results for legislators’ change in roll-call voting patterns: panels (1), (3), and (4) show changes in roll-call voting extremism, voting in line with the party majority, and voting in line with party leaders, respectively. In all three, there appears to be a negligible shift in the trend at the threshold, suggesting that primary election outcomes, and thus eligibility for the general election, do not significantly shift legislators’ post-primary policy positions.

Table 4 presents the formal estimates, and the results from both designs consistently

**Figure 5:** Regression Discontinuity Results on Legislators' Policy Positions



*Note:* Legislators who narrowly win and lose do not show a statistically significant difference in their changes in roll-call voting after the primaries. In all measures, the changes for both winners and losers are statistically indistinguishable from zero, implying that neither group significantly alters their policy positions. Each panel plots the change in the score itself. The proportional margin used for the effort activities has no counterpart here: a position has no volume to normalize, and an extremity score is measured from a conventional origin rather than a true zero, so a share or a percentage change of it carries no meaning.

indicate negligible changes in policy positions. Across all four columns, the regression-discontinuity coefficients are at most 0.01 in magnitude, and the difference-in-differences estimates are likewise small, suggesting minimal differences in how winners and losers adjust their positions. The table also reports the mean changes in policy positions for winners and losers separately, and in every case, these mean changes are close to zero. This pattern indicates not only that winners and losers change their behavior similarly after primaries,

but that neither group meaningfully alters its policy positions between pre- and post-primary periods.

These results suggest limited effects of electoral concerns on legislators’ ideological positions, contrasting with the conventional view of position-taking as central to the electoral connection and competition (Mayhew, 2004; Downs, 1965). The null instead aligns with empirical evidence of rigidity in candidate positions (Hall, 2019; Rogers, 2017; Poole and Rosenthal, 2014; Lee, Moretti and Butler, 2004) and formal models that assume fixed policy positions (Alesina, 1988; Osborne and Slivinski, 1996).

One might worry that a null on the average shift masks offsetting individual-level movement: legislators within the same party could move in opposing ideological directions after primaries, leaving the aggregate unchanged even as individuals move substantially. The natural test is to examine absolute change in roll-call extremism, which captures movement in either direction without cancellation. Graph (2) of Figure 5 and the second column of Table 4 show a slight increase in absolute repositioning among primary losers on this measure, significant under the difference-in-differences design but not the regression discontinuity, offering weak evidence that losers change their voting patterns somewhat more than winners after primaries. These patterns are consistent with the last-period (shirking) literature, in which legislators no longer subject to a reelection constraint vote closer to their own ideological preferences (Kalt and Zupan, 1990; Zupan, 1990; Bender and Lott, 1996). The statistical significance is not robust across the two designs, however, so this interpretation remains speculative. The magnitude speaks to the original worry—that sizable individual shifts might be cancelling out in the average. Taken at face value, losers’ absolute repositioning exceeds winners’ by 0.014, on a winner base of 0.032. That excess amounts to about one additional vote in seventy switching relative to the party pole; even at its upper confidence bound, about one in forty. Repositioning large enough to hide behind a flat average would have appeared here, and it does not.

The null on policy positions itself, by contrast, is highly robust: the same result holds

**Table 4:** Numerical Results on Change in Legislators' Policy Positions

| Regression Discontinuity |                        |                                 |                                         |                                        |
|--------------------------|------------------------|---------------------------------|-----------------------------------------|----------------------------------------|
|                          | Change in<br>Extremism | Absolute Change<br>in Extremism | Change in Voting<br>with Party Majority | Change in Voting<br>with Party Leaders |
| Primary Loss             | −0.003<br>(0.009)      | 0.008<br>(0.006)                | −0.003<br>(0.008)                       | 0.000<br>(0.008)                       |
| Observations             | 918                    | 918                             | 918                                     | 911                                    |
| Winner Mean              | 0.001                  | 0.037                           | 0.001                                   | 0.000                                  |
| Loser Mean               | 0.007                  | 0.048                           | −0.007                                  | −0.003                                 |
| Bandwidth                | 0.2                    | 0.2                             | 0.2                                     | 0.2                                    |
| Kernel                   | Uniform                | Uniform                         | Uniform                                 | Uniform                                |
| Specification            | Linear                 | Linear                          | Linear                                  | Linear                                 |

| Difference in Differences |                                     |                                 |                                         |                                        |
|---------------------------|-------------------------------------|---------------------------------|-----------------------------------------|----------------------------------------|
|                           | Change in<br>Extremism              | Absolute Change<br>in Extremism | Change in Voting<br>with Party Majority | Change in Voting<br>with Party Leaders |
| Primary Loss              | 0.006<br>(0.004)                    | 0.014**<br>(0.006)              | 0.002<br>(0.004)                        | 0.006<br>(0.004)                       |
| Observations              | 3,296,336                           | 3,694                           | 3,290,805                               | 2,892,750                              |
| Winner Mean               | −0.000                              | 0.032                           | −0.000                                  | 0.001                                  |
| Loser Mean                | 0.007                               | 0.047                           | −0.007                                  | −0.003                                 |
| Unit of Analysis          | Legislator–vote                     | Legislator–Congress             | Legislator–vote                         | Legislator–vote                        |
| Fixed Effects             | Legislator,<br>Bill-Party-<br>Cong. | Legislator,<br>Party-Congress   | Legislator,<br>Bill-Party-<br>Cong.     | Legislator,<br>Bill-Party-<br>Cong.    |

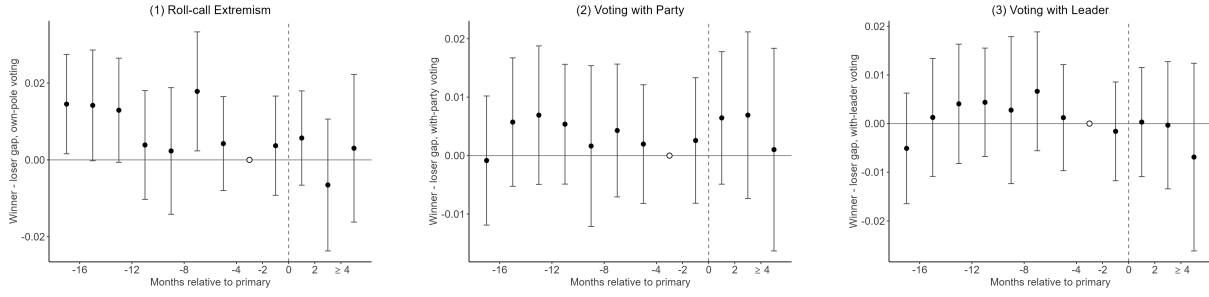
\* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$

*Note:* Coefficients are expressed as the effect of losing the primary. In the difference-in-differences panel, the “Primary Loss” row reports the  $\text{PrimaryLoss} \times \text{AfterPrimary}$  interaction (equation (4); legislator and bill-by-party-by-Congress fixed effects) for the three vote-level columns and the Primary Loss coefficient (aggregated legislator-period specification with legislator and party-Congress fixed effects) for the absolute-change column, which has no vote-level analog; with legislator fixed effects, that coefficient is identified within legislators whose primary outcome varies across their terms, not by a cross-sectional comparison of winners with losers. Winner Mean and Loser Mean give the average within-term change for terms won versus lost at the legislator–Congress level, computed on the  $\pm 0.2$  estimation sample in the regression-discontinuity panel and on all contested primaries in the difference-in-differences panel (0 indicates no change). For the extremism columns the dependent variable is the own-party-pole indicator  $\text{extreme}_{iv}$ . Difference-in-differences standard errors are clustered by legislator; regression-discontinuity standard errors are conventional OLS standard errors.

whether the dependent variable is the within-term regression coefficient  $\beta_{it}$  from equation (2) (RDD columns of Table 4), the vote-level own-party-pole, with-party, or with-leader indicators with roughly 2.9–3.3 million bill-vote observations each (DiD columns of Table 4), or a simple pre/post mean difference computed over equal-length windows without any regression adjustment (Table A.11). Nor is the null an artifact of imprecision: the estimates are precise enough that a with-party or with-leader shift of two votes in a hundred would have been detected with conventional power, so the data affirmatively bound any repositioning below that mark (Table F.1). This is the standard applied to null results throughout the paper: a null carries evidential weight only where the design could have detected an effect of the size at issue. The position nulls meet that standard and are read as evidence of absence. Nulls from designs without that power—the raw-count level estimates at the cutoff, whose standard errors are an order of magnitude larger than the changes at issue (Table A.3)—are read as absence of evidence, and no conclusion in the paper rests on them. A final check for the RDD asks whether the null change could conceal a winner–loser gap that is already present before the primary and simply persists afterward, since a constant gap differences away to zero. I estimate separate RDDs on the pre-primary and the post-primary voting levels (subsection A.2): at the cutoff, the pre- and post-primary levels are small and at most marginally significant for both with-party and with-leader voting. There is no meaningful gap before the primary and none after, so the null reflects genuine stability rather than a constant winner–loser gap that the change would difference away. The same roll-call data, by contrast, yields a statistically precise and substantively large effect on the effort margin of participation, indicating the null on policy positions reflects actual stability in legislators’ within-term voting rather than insensitivity of the measures.

The stability also has no hidden time structure. Figure 6 applies the same event-time lens as Figure 4 to the three vote-based position measures: the winner–loser gaps are flat throughout, before and after the primary alike, where participation breaks discretely at the primary date. The joint tests confirm what the panels show—post-primary gaps are jointly

**Figure 6:** Event-Time Stability of Policy Positions around the Primary



*Note:* Each panel traces the winner–loser gap in a vote-based position measure by event time relative to each legislator’s own primary date, with 95% confidence intervals and the open circle marking the omitted reference bin; the specification is identical to the participation event study of Figure 4 (close primaries, two-month bins, post-primary window ending at each legislator’s general election), estimated on cast votes for legislators with at least ten cast votes on each side of the primary. Estimates are plotted at the centers of their two-month windows, the dashed line at the primary date. The final bin pools weeks 16 and beyond: later event times exist only for legislators with early primaries, and the weeks-32-and-beyond tail holds only eight legislators in this sample, two of them losers, too few to estimate separately. Joint tests: pre-primary leads are flat for with-party ( $p = 0.90$ ) and with-leader voting ( $p = 0.75$ ) and marginally non-flat for extremism ( $p = 0.04$ ), driven by bins more than six months before the primary; post-primary gaps are jointly null at  $p = 0.47$ ,  $0.59$ , and  $0.88$  for extremism, with-party, and with-leader voting. The term-level tests of Table 4 carry the precision of the stability claim; bin-level confidence intervals are wider by construction.

null for all three measures. The far leads of the extremism measure are marginally non-flat ( $p = 0.04$ ), echoing the small pre-primary own-pole gap already documented in the level decomposition (subsection A.2); the bins adjacent to the primary are null. The precision of the stability claim rests on the term-level estimates and equivalence bounds above; the event studies add that the term-level null is not averaging away a short-lived repositioning around the primary.

## 6 Heterogeneous Effects of Electoral Incentives

### 6.1 How Effort Effects Vary with General-Election Competitiveness

If legislators who lose their primaries withdraw effort, a natural reading is that the forgone post-primary effort was strategic preparation for the upcoming general election, where such activity can enhance a candidate’s vote share. In some districts, however, the general election is expected to be minimally competitive, especially when a legislator or their party holds a dominant position. In such districts, legislators may worry less about reelection, and the reelection incentive that a primary loss removes may be correspondingly weaker (Burden, 2004). This suggests that the effort effect could depend on anticipated competition in the general election.

To test this, I divide the sample by whether the upcoming general election is expected to be competitive. Because incumbents in uncompetitive districts typically win by large margins, I use the incumbent’s vote share in the most recent general election: districts where the incumbent received more than 75% form the uncompetitive group, and those below form the competitive group.

Table 5 shows that the withdrawal is general rather than confined to competitive environments. Roll-call participation falls in both district types under both designs. Constituent communication—the most campaign-facing of the four activities—falls in safe districts as well: losers there send 0.22 fewer e-newsletters per week on the weekly-panel specification of Table 3, a decline that is stable at every competitiveness cutoff between 0.60 and 0.75 and survives a wild cluster bootstrap over the ten losing terms that identify it ( $p = 0.005$ ). Behind the coefficient stands a simpler fact: the eight of those losers who used the medium at all sent 227 newsletters before their primaries and 4 after. Cosponsorship shows nearly identical proportional-margin estimates in the two district types ( $-0.114$  and  $-0.115$ ), though the smaller safe-district sample renders the latter marginal ( $p = 0.06$ ), and its uncompetitive

**Table 5:** Effort Effects by General-Election Competitiveness: Two Designs, Two Margins

| Competitive Districts ( $V_{t-1} < 0.75$ )                                  |                                                     |                                                    |                                                      |                                   |
|-----------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|------------------------------------------------------|-----------------------------------|
|                                                                             | Roll-call<br>Participation<br>(share of votes cast) | Bill<br>Sponsorship<br>(count per<br>session-week) | Bill<br>Cosponsorship<br>(count per<br>session-week) | E-Newsletters<br>(count per week) |
| <i>Additive margin: change in rate</i>                                      |                                                     |                                                    |                                                      |                                   |
| Primary Loss (RD)                                                           | -0.088***<br>(0.023)                                | -0.051<br>(0.076)                                  | -1.632***<br>(0.534)                                 | -0.914***<br>(0.233)              |
| Observations                                                                | 660                                                 | 487                                                | 424                                                  | 179                               |
| Primary Loss ×<br>After (DiD)                                               | -0.110***<br>(0.015)                                | -0.063***<br>(0.021)                               | -1.072***<br>(0.162)                                 | -0.803***<br>(0.258)              |
| Observations                                                                | 2,567,939                                           | 171,634                                            | 149,164                                              | 69,497                            |
| <i>Proportional margin: share of activity / change relative to own rate</i> |                                                     |                                                    |                                                      |                                   |
| Primary Loss (RD,<br>proportion)                                            | -0.040<br>(0.032)                                   | -0.149*<br>(0.077)                                 | -0.114***<br>(0.035)                                 | -0.219***<br>(0.082)              |
| Observations                                                                | 682                                                 | 417                                                | 421                                                  | 155                               |
| Primary Loss ×<br>After (Poisson)                                           | -0.131***<br>(0.020)                                | -0.608***<br>(0.181)                               | -0.415***<br>(0.053)                                 | -1.170***<br>(0.244)              |
| Observations                                                                | 2,567,939                                           | 161,502                                            | 150,720                                              | 59,748                            |
| Uncompetitive Districts ( $V_{t-1} > 0.75$ )                                |                                                     |                                                    |                                                      |                                   |
|                                                                             | Participation                                       | Sponsorship                                        | Cosponsorship                                        | E-Newsletters                     |
| <i>Additive margin: change in rate</i>                                      |                                                     |                                                    |                                                      |                                   |
| Primary Loss (RD)                                                           | -0.199***<br>(0.039)                                | 0.042<br>(0.205)                                   | -0.791<br>(1.519)                                    | 0.230<br>(0.476)                  |
| Observations                                                                | 266                                                 | 178                                                | 154                                                  | 66                                |
| Primary Loss ×<br>After (DiD)                                               | -0.085**<br>(0.035)                                 | 0.007<br>(0.071)                                   | -0.754<br>(0.499)                                    | -0.224***<br>(0.070)              |
| Observations                                                                | 952,477                                             | 64,791                                             | 57,132                                               | 26,566                            |
| <i>Proportional margin: share of activity / change relative to own rate</i> |                                                     |                                                    |                                                      |                                   |
| Primary Loss (RD,<br>proportion)                                            | -0.134**<br>(0.060)                                 | -0.206<br>(0.161)                                  | -0.115*<br>(0.060)                                   | 0.327*<br>(0.190)                 |
| Observations                                                                | 278                                                 | 147                                                | 154                                                  | 55                                |
| Primary Loss ×<br>After (Poisson)                                           | -0.101**<br>(0.044)                                 | 0.136<br>(0.385)                                   | -0.291*<br>(0.160)                                   | -2.240***<br>(0.580)              |
| Observations                                                                | 949,426                                             | 55,913                                             | 57,400                                               | 21,861                            |

\* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ 

*Note:* Coefficients are the effect of losing the primary; standard errors are conventional in the discontinuity rows, clustered by legislator in the panel rows. Panels split districts at 75% of the incumbent's most recent general-election vote share. Margins follow Table 3: additive rows are changes in weekly rates under that table's measurement convention (session-week rates and session-week panels for sponsorship and cosponsorship), with difference-in-differences rows estimated on Table 3's weekly-panel specification; proportional rows are post-primary shares of activity at the discontinuity and fixed-effects Poisson estimates in the panel; the participation proportional cells translate the additive response through the base rate, as in Table 3. Count-panel subgroup samples omit terms without a recorded previous vote share and one 116th-Congress party switcher whose mid-term party coding is undefined; full-sample estimates are unchanged at printed precision. Sixteen formal gradient tests—continuous and cutoff interactions across all outcomes, margins, and designs—are null at the five-percent level. The uncompetitive sponsorship and cosponsorship panel nulls are uninformative rather than contrary: their minimum detectable effects (1.08 proportional, 1.40 additive) exceed the corresponding full-sample estimates. The uncompetitive e-newsletter panel estimate rests on ten losing terms, eight with any newsletters (227 before the primary, 4 after); it is stable at every cutoff between 0.60 and 0.75 and survives a wild cluster bootstrap over those terms ( $p = 0.005$ ; randomization inference  $p = 0.08$  at this cut,  $\leq 0.03$  at every broader one). The uncompetitive e-newsletter proportional cells rest on the same eight legislators and disagree in sign across implementations, describing those legislators rather than supporting an inference.

additive cell is too imprecise to be informative. Sponsorship is detected only in competitive districts; its uncompetitive cells cannot distinguish an equal effect from none at all, with a minimum detectable effect on the proportional margin more than twice the full-sample estimate.

Formal tests agree with what the table shows: across sixteen interaction and cutoff tests—every outcome, both margins, both designs—none detects a competitiveness gradient at the five-percent level.<sup>5</sup> The point estimates leave room for the communication response to be larger where campaigns loom larger—the e-newsletter decline is roughly three and a half times the size in competitive districts (0.80 versus 0.22 newsletters per week), and sponsorship registers only there—but the data do not order these magnitudes statistically, and for participation the point pattern runs, if anything, the other way. What the design does establish is presence: wherever it has the power to look, in the safest districts and the most contested alike, losing the primary is followed by withdrawal.

Notably, the regression-discontinuity participation decline is larger in uncompetitive districts ( $-0.199$ , about twenty percentage points of participation) than in competitive ones ( $-0.088$ ). This pattern is specific to the regression discontinuity, however: in the more highly powered difference-in-differences design the two are statistically indistinguishable ( $-0.110$  versus  $-0.085$ ), and the uncompetitive regression-discontinuity estimate rests on a small close-primary sample of 266 legislators. I therefore read it cautiously. If it reflects a real difference, a plausible account is that a substantial share of legislators’ motivation to attend roll-calls is non-electoral, such as personal ideological commitments, consistent with the earlier finding that losers still take part in most roll-calls, albeit less frequently than winners; on this account, primary winners in competitive districts attend somewhat fewer roll-calls because they must devote more time to campaigning and visiting home districts ahead of the general election. But because the two designs disagree, and a formal interaction test

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<sup>5</sup>One of the sixteen reaches the ten-percent level (the e-newsletter volume interaction,  $p = 0.059$ , pointing toward a larger response in competitive districts)—within what chance alone would produce across sixteen tests. The tests also have limited power against moderate gradients: they establish that the withdrawal is present in both district types, not that its magnitude is everywhere equal.

detects no gradient ( $p = 0.31$ ), the safer conclusion is that the participation effect is broadly comparable across district types.

The pattern speaks to what the incentive is. If effort tracked the marginal electoral return of the upcoming general election, safe-district losers—whose defeat changes little about a foregone conclusion—should barely respond; instead they withdraw as sharply as anyone. What loss removes in both district types alike is candidacy itself: the extensive margin of having a next election at all, not the intensive margin of how close it will be. Nor does the withdrawal track the length of the career ahead: age and seniority splits are uniform—losers below the median age of 55, for whom careers beyond the House loom largest, withdraw no less than their older colleagues (Appendix A.17)—which tells against accounts in which effort protects a remaining career rather than a candidacy. One alternative the heterogeneity cannot reach remains: because a primary defeat ends the House career along with the campaign, the design cannot separate the electoral incentive from the prospect of another term—for House members the two coincide. The timing established earlier answers a narrower question, whether the withdrawal is a career winding down gradually as it ends. It is not. Effort remains flat through the final year and breaks discretely at each legislator’s own primary date. A career winding down would not produce that path.

## **6.2 Effects on Policy Positions Remain Null for Important Bills**

The null on policy positions might stem from a narrower channel: legislators may respond to electoral incentives only on a small subset of salient or important bills. To test this, I restrict the analysis to bills deemed important.

I take three approaches to identifying “important” bills. First, I examine bills “whipped” by party leadership to test whether legislators respond to party pressure. Party leaders occasionally conduct whip counts — informal surveys on significant agenda items — typically reserved for high-priority bills. If electoral incentives operate through party pressure, the effects should be more pronounced on whipped bills. This analysis leverages whip count data

from the Democratic Party (1955–1986) and the Republican Party (1975–1980) provided by Evans (2012).

Second, I focus on bills previous scholars deemed especially significant or salient. I pool two standard measures of legislative importance, each linked to specific bill numbers: priority agenda items from Curry and Lee (2020) and landmark enactments compiled from CQ Press.<sup>6</sup> I apply the same measure of position change to these bills. If legislators are more responsive to high-profile legislation, the effects should be more pronounced on them than across all bills.

Third, I focus on contested bills — those close to passing or failing. Such bills are especially informative because each legislator is more likely to cast a pivotal vote. If electoral incentives heighten legislators’ attention to legislative outcomes, the effects should be more pronounced on these narrowly decided bills. To operationalize this, I calculate the difference between yea and nay votes for each bill and repeat the analysis for bills in the lowest 30% of that difference.<sup>7</sup>

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<sup>6</sup>The priority-agenda series covers only Congresses 103–115 and supplies too few within-window votes per legislator to support the regression-discontinuity change-score once the minimum-vote requirement below is imposed; the landmark series spans Congresses 84–113 and supplies the needed coverage. The two measures are largely redundant, so the results are nearly identical using the landmark series alone, while the priority-agenda series alone is underpowered.

<sup>7</sup>30% is arbitrary; results are unchanged at the 20% or 50% thresholds. This measure of “contestedness” is posterior, capturing realized vote outcomes after electoral incentives may have influenced legislators’ votes, but realized contestedness likely approximates expected contestedness.

**Table 6:** Effects of Primary Loss on Voting on Important Bills

|                                         | Regression Discontinuity                  |                                   |                      | Difference-in-Differences           |                                   |                      |
|-----------------------------------------|-------------------------------------------|-----------------------------------|----------------------|-------------------------------------|-----------------------------------|----------------------|
|                                         | Whipped<br>Bills                          | Significant/<br>Landmark<br>Bills | Contested<br>Bills   | Whipped<br>Bills                    | Significant/<br>Landmark<br>Bills | Contested<br>Bills   |
| Change in<br>Roll-call Extremism        | 0.051<br>(0.075)                          | 0.013<br>(0.022)                  | 0.000<br>(0.016)     | 0.072<br>(0.048)                    | 0.015**<br>(0.008)                | 0.010<br>(0.006)     |
| Change in Voting<br>with Party Majority | 0.046<br>(0.081)                          | 0.012<br>(0.021)                  | −0.010<br>(0.016)    | 0.029<br>(0.047)                    | 0.014*<br>(0.008)                 | 0.010<br>(0.006)     |
| Change in Voting<br>with Party Leaders  | 0.137<br>(0.086)                          | 0.000<br>(0.023)                  | 0.004<br>(0.018)     | 0.078<br>(0.049)                    | 0.018*<br>(0.009)                 | 0.015**<br>(0.007)   |
| Change in Roll-call<br>Participation    | −0.118**<br>(0.052)                       | −0.107***<br>(0.027)              | −0.112***<br>(0.020) | −0.161***<br>(0.038)                | −0.086***<br>(0.018)              | −0.099***<br>(0.013) |
| Observations                            | 124–136                                   | 656–684                           | 899–909              | 28k–36k                             | 496k–623k                         | 942k–<br>1,101k      |
| Specification                           | Bandwidth = 0.2<br>Specification = Linear |                                   |                      | Legislator + Bill-Party-Congress FE |                                   |                      |

\* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$

*Note:* Each cell reports the effect of losing the primary on the change in the respective voting behavior; standard errors are in parentheses and clustered by legislator in the difference-in-differences design. The difference-in-differences and regression-discontinuity specifications are the same as in Table 3, and the roll-call extremism outcome is the own-party-pole indicator  $extreme_{iv}$  defined in subsection 3.1; the with-party and with-leader outcomes are the corresponding vote-level indicators, so coefficients are changes in the share of votes. The three bill subsamples are whipped bills (Evans 2012; Congresses 84–99), significant/landmark bills (priority agenda items from Curry & Lee 2020 pooled with landmark enactments compiled from CQ Press; Congresses 84–115), and contested bills (bottom 30% of  $|yea - nay|$ ; Congresses 84–117). To estimate a legislator’s within-subsample position or participation change, the regression-discontinuity columns require at least five subsample votes on each side of the primary (cast votes for the position outcomes, available roll calls for participation; Appendix A.6); the difference-in-differences columns pool all votes and impose no such requirement. Regression-discontinuity sample sizes therefore differ across rows (positions versus participation) and are reported as ranges; vote-level sizes likewise differ within each column due to outcome-specific missingness. Because the difference-in-differences samples for the significant/landmark and contested columns exceed half a million votes, coefficients there can be statistically distinguishable from zero while remaining substantively negligible in magnitude ( $|\hat{\beta}| < 0.02$ ).

Each cell in Table 6 reports the effect of losing a primary on subsequent voting for whipped, significant/landmark, and contested bills. Across both designs, coefficients for roll-call extremism and voting alignment (with party majority or party leaders) are small

and substantively indistinguishable from zero. A few of the difference-in-differences position estimates reach conventional significance in the significant/landmark and contested columns, but only because those pooled vote-level samples exceed half a million observations; the coefficients themselves are negligible, all below 0.02 in absolute value. This pattern reinforces the earlier evidence on the rigidity of legislators’ policy positions: primary winners and losers do not systematically differ in how they adjust positions on major votes after primary elections. The bottom row provides a power check. On all three subsamples—each restricted to legislators with at least five available roll calls per side, so the change-scores are well estimated—roll-call participation falls sharply and significantly for primary losers, so the flat policy positions reflect genuine stability rather than a lack of statistical power. Nor does the null stem from losers disengaging—their participation falls only about nine percentage points after the primary while winners’ is essentially unchanged (Table 3, note), leaving them active on a substantial share of votes.

## 7 Conclusion

Electoral accountability lies at the heart of democratic governance, compelling representatives to remain responsive to constituents. Alexander Hamilton underscored this during the debates over term limits, arguing that removing the prospect of reelection would diminish legislators’ motivation: “One ill effect of the exclusion [of reelection] would be a diminution of the inducements to good behavior” (Hamilton, 2003).

Despite the recognized importance of electoral incentives, identifying precisely which “good behavior” they induce remains empirically challenging. Representatives strategically choose whether to seek reelection or retire, making it difficult to isolate the influence of electoral incentives from selection bias. This study addresses this challenge by leveraging the post-primary period, when incumbents face a critical juncture: continuing their reelection campaigns or exiting after a primary defeat. Legislators who lose their primaries withdraw

legislative effort: roll-call participation falls sharply, in a decline timed to the primary itself, and bill sponsorship, cosponsorship, and constituent communication fall in step, each by roughly a third or more of a legislator’s own rate. Yet legislators’ policy positions in roll-call voting do not significantly change, whether they won the primary or not.

Taken together, the results show that electoral incentives discipline legislative effort but not legislative position-taking. The effort findings reinforce classic theories of electoral accountability (Barro, 1973; Ferejohn, 1986; Mayhew, 2004): legislators sustain floor participation, bill sponsorship and cosponsorship, and constituent communication so long as they remain candidates for another term—and they do so in safe districts and competitive ones alike, whatever the length of the career ahead. Contrary to the worry that voters’ inattention blunts accountability (Achen and Bartels, 2017), what disciplines representatives is not the intensity of the contest but the prospect of facing voters at all.

Their roll-call positions, by contrast, barely change. Although position-taking is classically treated as central to the electoral connection (Mayhew, 2004) and electoral competition (Downs, 1965), positions hold steady after a primary defeat, even when electoral pressure should produce the sharpest response: whipped bills, significant agenda items, and narrowly contested ones. The estimates are precise enough to detect a party-line shift as small as two votes in a hundred, so the stability is affirmatively bounded, not merely undetected. I interpret this divergence as reflecting a difference in kind between the two margins. Effort is an individually controlled, observable signal that can pay off in the next campaign; roll-call positions are anchored by more durable commitments—members’ own ideological convictions and the established policy records they were elected on (Poole and Rosenthal, 2014; Hall, 2019; Lee et al., 2004; Alesina, 1988; Osborne and Slivinski, 1996)—that the lifting of reelection pressure does not dislodge.

The same stability informs how scholars read roll-call data. Parties control resources legislators need to win (Cox and McCubbins, 2007; Krehbiel, 1993), yet party-line voting is unchanged when a member’s reelection incentive vanishes, so within-term electoral pressure

is not the primary channel through which parties pull members into line. For the same reason, because ideological scores such as DW-NOMINATE (Poole and Rosenthal, 1985; McCarty, 2016) are stable across the primary cutoff even on the strongest-prediction bills, they are unlikely to be mere artifacts of immediate electoral pressure operating through party discipline or donors (Snyder and Groseclose, 2000; Bronars and Lott, 1997).

The findings also bear on the roots of polarization. Because sitting incumbents do not shift their positions in their term’s closing months, regardless of primary outcome, widening partisan divides are more plausibly driven by who enters and leaves office—replacement, sorting, and nomination—than by repositioning among those already in office (Moskowitz et al., 2024).

This null is subject to one qualification: it describes the months after a primary, not elections in general. The design isolates the within-term effect of removing future reelection incentives; it does not preclude electoral incentives from shaping policy at a slower pace over a longer time span. Albouy (2011) finds that U.S. Senators, who serve six-year terms, moderate their voting more in the years closer to reelection, and Caughey and Warshaw (2018) document multi-year responsiveness of state policy to public opinion—evidence that electoral pressure may accumulate over a longer horizon. A null effect over a relatively short period in this design does not necessarily contradict accounts in which electoral pressure builds slowly and positions, once taken, are costly to abandon, so that its abrupt removal over a few months leaves recorded votes unchanged. Nor can the design detect positioning that adjusted *before* the primary—a member casting tough votes in anticipation of not running again—or change deterred by the very threat of a challenge; the null accordingly bounds the post-primary, within-term channel rather than every path by which electoral competition could discipline positions.

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